



# PhotonWorks

Astrophotography stacking and image processing

## User Manual

Version 1.0.9

PhotonWorks is a Windows application for stacking and processing astrophotography. It bundles the Siril engine for calibration, registration and stacking, and provides its own image editor and 41 processing tools built natively in the app.

Copyright (C) 2026 Stewart Oliver. Released under the GNU General Public License v3.

# Contents

---

|                              |           |
|------------------------------|-----------|
| <b>Before you start</b>      | <b>4</b>  |
| What it is                   | 4         |
| What you need                | 4         |
| The first run                | 4         |
| Finding your way around      | 5         |
| <b>Loading your frames</b>   | <b>6</b>  |
| The frame list               | 6         |
| <b>Stacking</b>              | <b>8</b>  |
| Core Parameters              | 8         |
| Stacking methods             | 8         |
| Running it                   | 8         |
| <b>The Image Editor</b>      | <b>9</b>  |
| Opening several at once      | 9         |
| Windows                      | 9         |
| Icons                        | 10        |
| Arranging them all at once   | 11        |
| More room when you need it   | 12        |
| The editor toolbar           | 12        |
| <b>How editing works</b>     | <b>13</b> |
| Preview, then apply          | 13        |
| Undo and redo                | 13        |
| Saving                       | 13        |
| <b>Masks</b>                 | <b>14</b> |
| <b>Tool reference</b>        | <b>15</b> |
| Masking (2)                  | 15        |
| Image Prep (5)               | 16        |
| Calibration & Astrometry (3) | 20        |
| Background Tools (4)         | 23        |
| Noise & Sharpening (1)       | 27        |
| Star Tools (5)               | 28        |
| Stretch Tools (5)            | 33        |
| Tone (1)                     | 38        |
| Structure & Detail (3)       | 39        |
| Colour Tools (5)             | 42        |

|                                     |           |
|-------------------------------------|-----------|
| Channels & Palettes (7)             | 48        |
| <b>A worked example</b>             | <b>55</b> |
| <b>Settings</b>                     | <b>57</b> |
| Scratch space                       | 57        |
| Interface size                      | 58        |
| Updates                             | 58        |
| <b>If something goes wrong</b>      | <b>59</b> |
| <b>Credits and licensing</b>        | <b>60</b> |
| Licence                             | 61        |
| <b>Menus and keyboard shortcuts</b> | <b>62</b> |
| <b>Version history</b>              | <b>64</b> |

## CHAPTER 1

# Before you start

What PhotonWorks is, what it needs, and what happens the first time you run it.

## What it is

PhotonWorks takes the frames your camera produced and turns them into one finished image. It does that in two halves, and they are worth keeping apart in your head, because they work quite differently.

The first half is STACKING: calibrating your light frames against darks, flats and bias, registering them to each other, and combining them into a single master. That work is done by Siril, which is bundled inside the program - you do not install it, and there is nothing to configure. PhotonWorks writes the script, runs the engine, and reports back.

The second half is EDITING: stretching, colour, structure, stars, noise, palettes. Almost none of that is Siril. The tools in the right-hand sidebar - 41 of them across 11 groups - are written natively in the app and run on the image in the Editor. The handful that are exceptions say so in their own description: Background Extraction, SPCC and the Image Solver are Siril; GraXpert and the Xterminator tools are separate programs you install yourself and point the app at.

**Note.** The design rule the app is built to is 'Siril only for stacking, everything else built from scratch.' If a tool is not in that exceptions list, what it does is happening inside PhotonWorks, on your pixels, and Undo will take it back.

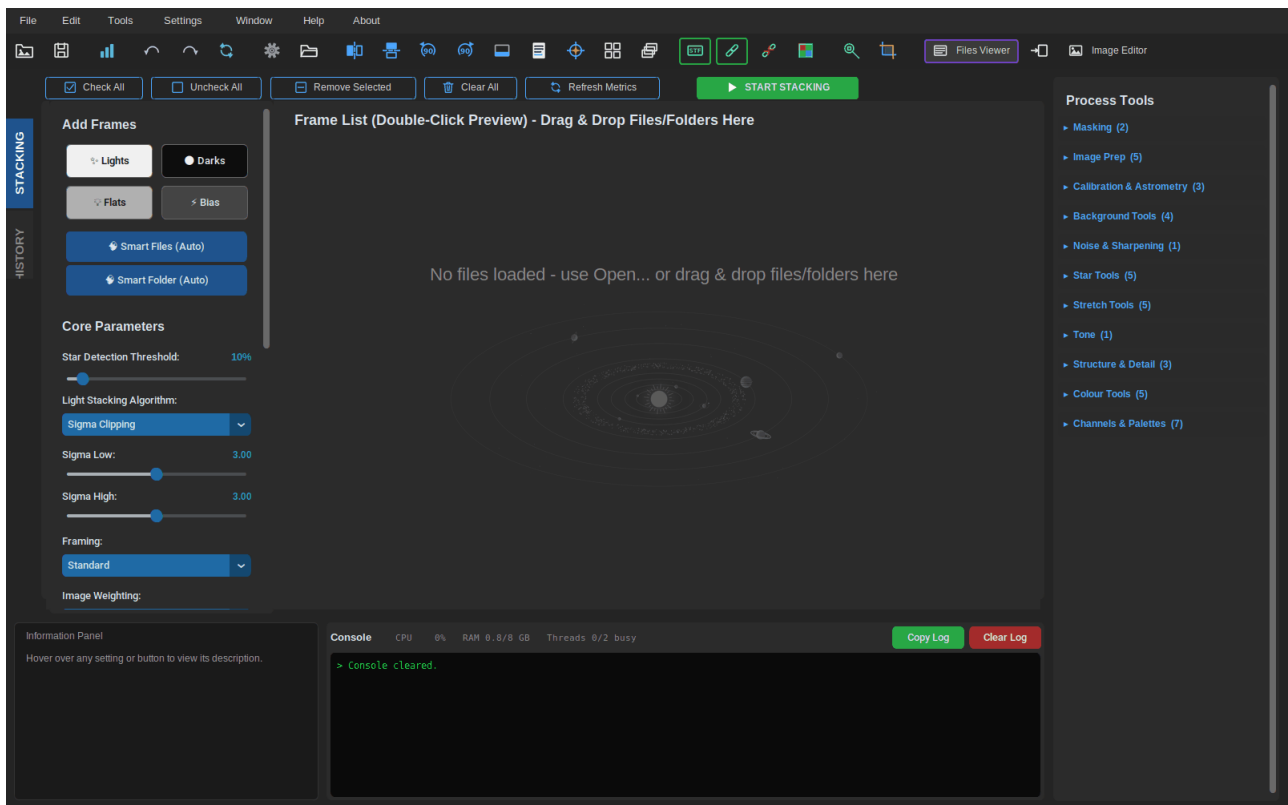
## What you need

|          |                                                                                                                                                                                                                               |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows  | PhotonWorks is a Windows application, shipped as an .exe with an _internal folder beside it. Keep the two together.                                                                                                           |
| Disk     | Stacking writes a lot of intermediate data. Several tens of gigabytes is normal for a night's worth of subs, and the scratch folder can be moved to another drive - see Chapter 10.                                           |
| Internet | Only for plate solving, SPCC and the update check. Everything else works offline.                                                                                                                                             |
| Optional | GraXpert, and the RC-Astro CLI for BlurXTerminator, StarXTerminator and NoiseXTerminator. Both are separate installs, and both are optional - the app runs without them and simply says so if you open a tool that needs one. |

## The first run

PhotonWorks opens on the stacking page. Nothing is loaded, and the large panel in the middle is where your frames will be listed.

If anything is wrong - a missing Siril engine, a path that no longer exists - the app says so in a notice at startup rather than failing later in the middle of a stacking run. The console at the bottom of the window carries a running commentary of everything the app and the engine do, and it is the first place to look when something does not behave.



The stacking page as it opens. Frames go in the middle, stacking parameters on the left, processing tools on the right, console along the bottom.

## Finding your way around

| Where       | What lives there                                                                                                                                                                                                   |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Left panel  | Add Frames buttons, and the Core Parameters that control the stack. Collapses with F9.                                                                                                                             |
| Centre      | Two views sharing one space: the Files Viewer, which is the frame list, and the Image Editor, which is where images are opened and worked on. The two buttons at the top right of the toolbar switch between them. |
| Right panel | Process Tools - the tool groups, as collapsible drawers. They all start closed, so the whole structure is visible at a glance.                                                                                     |
| Bottom      | The Information Panel, which describes whatever you hover over, and the Console. Both hide together with F8.                                                                                                       |
| Top         | The menu bar, and below it a single row of icons for the things you reach for constantly.                                                                                                                          |

## CHAPTER 2

# Loading your frames

---

Getting lights, darks, flats and bias into the frame list.

There are three ways in, and they differ only in how much sorting you do yourself.

### The four frame buttons

Lights, Darks, Flats and Bias each open a file browser and add whatever you pick as that type. This is the explicit route: you say what each file is, so nothing can be guessed wrongly.

### Smart Files and Smart Folder

Smart Files (Auto) takes a selection of files and works out what each one is for itself. Smart Folder (Auto) does the same for everything in a folder, including its subfolders. Both read the frame type out of each file's own header, so a set exported from your capture software normally sorts itself correctly in one go.

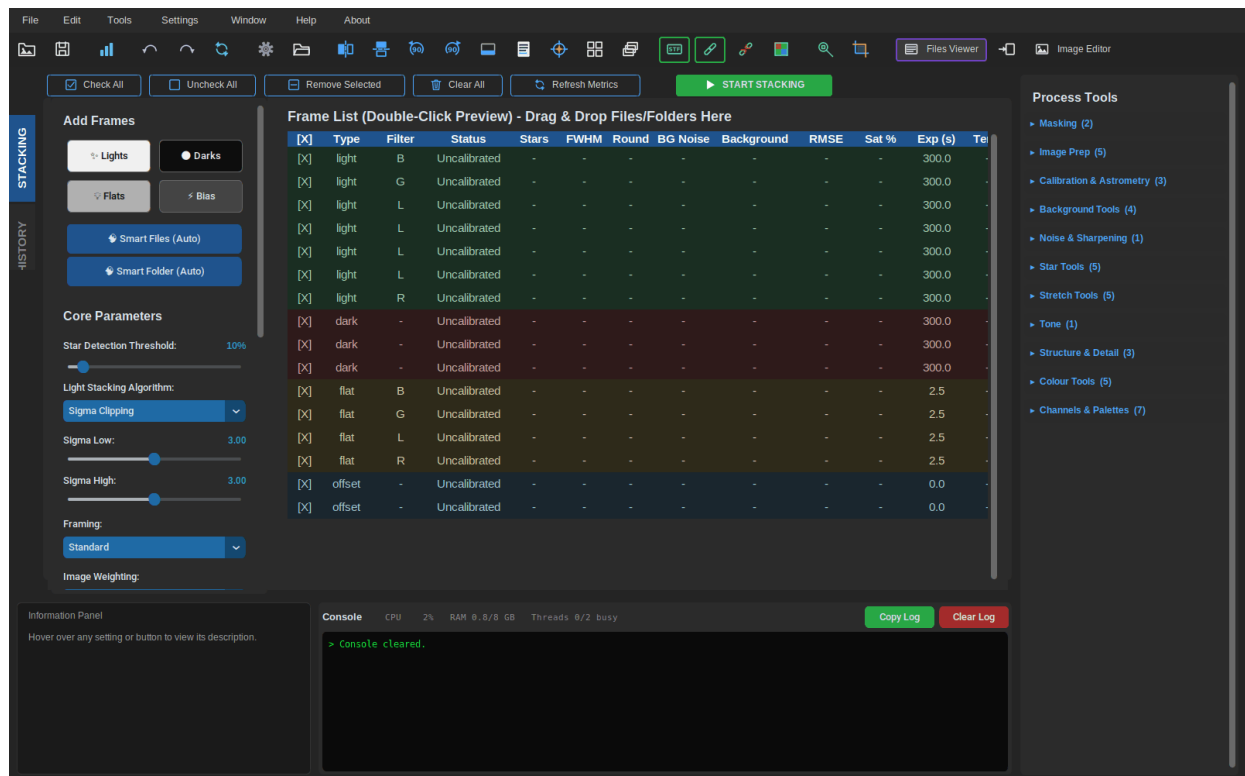
### Drag and drop

Files and folders can be dropped straight onto the frame list. The console says on startup whether drag and drop is available.

|                         |                                                                                                                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Formats it will read    | .arw, .cr2, .fit, .fits, .fts, .jpg, .nef, .png, .tif                                                                                                                                                |
| Already-stacked masters | A dark, flat or bias that is already a stacked master is recognised as one and used directly instead of being stacked again. The frame list marks it Master, and the console says how many it found. |

## The frame list

Every frame you add appears as a row, colour-coded by type. The columns to the right of the name are the per-frame measurements, and they stay empty until the frames have been registered - there is nothing to measure before then.



The Files Viewer with lights, darks, flats and bias loaded. Exposure, filter and sensor temperature are read from each file's own header; the measurement columns fill in after registration.

| Column                  | What it tells you                                                                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [X]                     | Whether the frame is included in the stack. Check All, Uncheck All and Remove Selected are above the list.                                                    |
| Type / Status           | Light, dark, flat or offset, and how far through the process it has got - Uncalibrated, Registered, Completed.                                                |
| Stars / FWHM / Round    | How many stars were detected, how big they are, and how circular. The three numbers that tell you whether a sub is sharp, and whether the mount was behaving. |
| BG Noise / Background   | The noise level and the sky level. A frame with a much higher background than its neighbours is usually cloud, moon or a passing light.                       |
| RMSE                    | Registration error - how well that frame lined up.                                                                                                            |
| Sat %                   | How much of the frame is saturated.                                                                                                                           |
| Exp (s) / Temp / Filter | Straight from the file header.                                                                                                                                |

Refresh Metrics (F5) re-reads the registration data and fills those columns in. Clear All empties the list entirely.

## CHAPTER 3

# Stacking

The parameters that matter, and what each of them changes.

The Core Parameters on the left are the settings you are most likely to change from one target to the next. Everything else lives in Advanced Settings, and sensible defaults mean you can leave most of it alone.

## Core Parameters

| Setting                  | What it does                                                                                                                                                                                   |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Star Detection Threshold | How faint a point of light has to be before registration stops counting it as a star. Raise it on a dense field, lower it if too few stars are being found to register at all.                 |
| Light Stacking Algorithm | How the frames are combined and, where it applies, how outliers are rejected. The full list is below.                                                                                          |
| Sigma Low / Sigma High   | How far from the average a pixel has to stray before it is thrown out - low for dark outliers, high for bright ones like satellite trails and cosmic rays. Only used by the rejection methods. |
| Framing                  | What the output frame covers: Standard, Mosaic, Auto-Crop (Intersection), or Centre of Gravity. Auto-Crop keeps only the area every frame has in common, so there are no ragged edges.         |
| Image Weighting          | Whether better frames count for more: None, FWHM (Sharpness), Number of Stars, Background Noise, or Frames Stacked.                                                                            |

## Stacking methods

Twelve entries, which are really two things: a combination method, and - for the mean-based ones - a rejection rule that decides which pixels get left out before the average is taken.

|                                                                           |
|---------------------------------------------------------------------------|
| Winsorized Clipping, Sigma Clipping, Median Clipping, Percentile Clipping |
| Linear Fit Clipping, GESD Clipping, MAD Clipping, No Rejection            |
| Median, Sum, Minimum, Maximum                                             |

**Note.** Winsorized Clipping is the default and the right answer most of the time. Sum is for planetary and lucky-imaging work, where every photon counts and there are no outliers to reject; Median is robust but noisier than a good rejected mean; Minimum and Maximum are diagnostic rather than everyday tools.

## Running it

START STACKING is the green button above the frame list. From there the console is the thing to watch: it carries the engine's own output line by line, so you can see calibration, registration and integration happen, and any error the engine reports appears there in full rather than as a summary.

Siril aborts a whole script at its first error. If a run stops early, the last few console lines before it stopped are the cause, not the last line printed.

When the run finishes, the result is written to the output folder and the frame list rows are marked Completed. If Auto-open is switched on - the small toggle between the two view buttons - the result opens straight into the Image Editor; otherwise double-click it in the Files Viewer when you are ready.

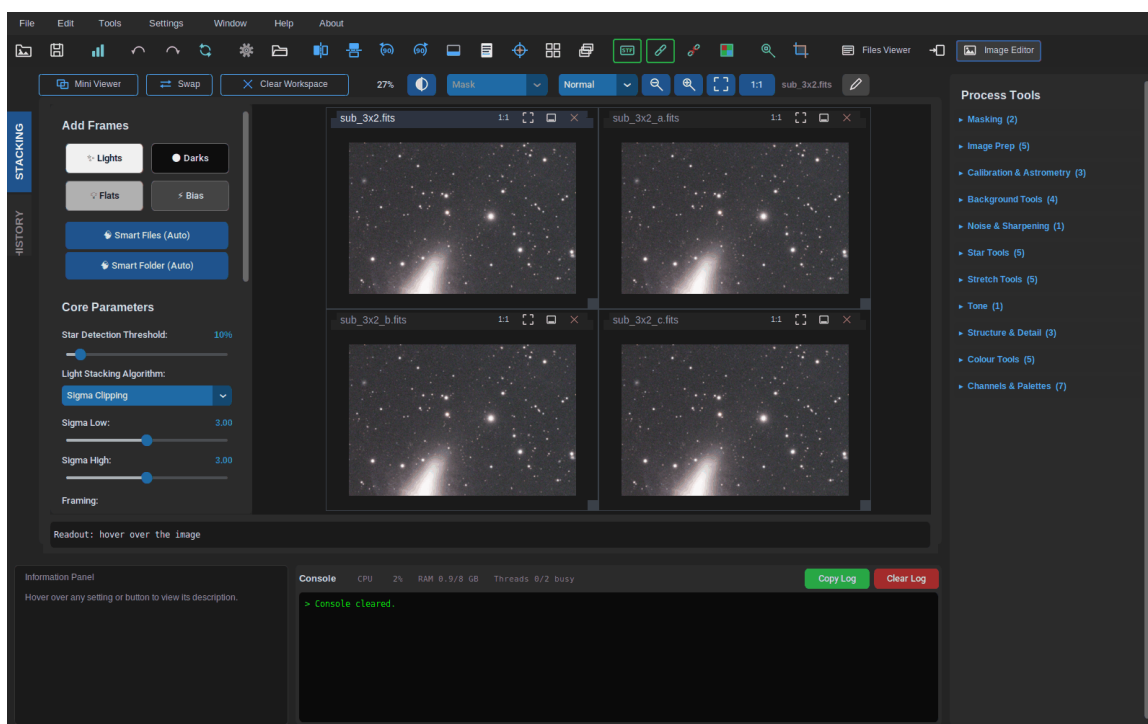
## CHAPTER 4

# The Image Editor

One workspace, as many images as you like, and how to arrange them.

The Image Editor is a workspace, not a single picture. Any number of images can be open at once. Each one keeps its own zoom, its own screen stretch and its own Undo history, so moving between them costs nothing and loses nothing - the image you come back to is exactly where you left it.

At any moment an image is in one of two states: it is a WINDOW on the workspace, or it is an ICON. One of the windows is the image being edited - its title bar is lighter, its name is in the toolbar, and it is the one every tool, Undo and Save will act on. Clicking another window hands editing to that one.



Four frames open at once. The lighter title bar, top left, is the image being edited.

## Opening several at once

Open... takes more than one file. Select four and all four come up together, tiled, with the first of them in the editor. Dragging several files onto the workspace does exactly the same thing - it is the same operation, just without the file dialog.

Images opened this way are windows straight away. Nothing has to be un-minimised or arranged by hand first.

## Windows

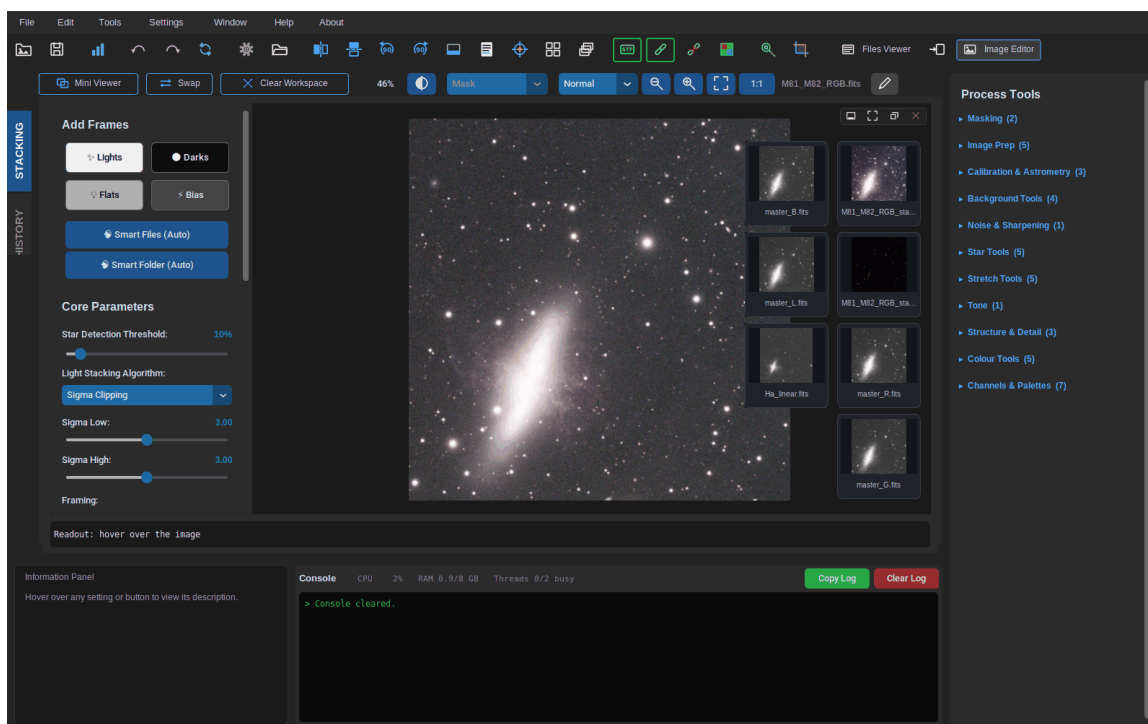
Each window has its own small title bar with four controls at the right-hand end:

| Control | What it does                                                                                   |
|---------|------------------------------------------------------------------------------------------------|
| 1:1     | Show this image at actual pixel size. The button is lit while that is what you are looking at. |
| Fit     | Fit the whole image into the window. Double-clicking the picture does the same.                |

| Control  | What it does                                                                                                                                                |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Minimise | Send this image down to an icon. Nothing is closed - its Undo history, stretch and zoom all survive, and double-clicking the icon brings it back as it was. |
| Close    | Take this image out of the app. Nothing is deleted from disk, and you are asked first if it is carrying unsaved edits.                                      |

Drag a window by its title bar to move it, or any edge or corner to resize it. Double-click the title bar to fill the editor with that image, and double-click again to bring it back to a window. Ctrl+Tab and Ctrl+Shift+Tab step through the windows in the order they were opened.

An image filling the editor has no title bar of its own - the same four controls appear in the top right corner of the workspace instead.

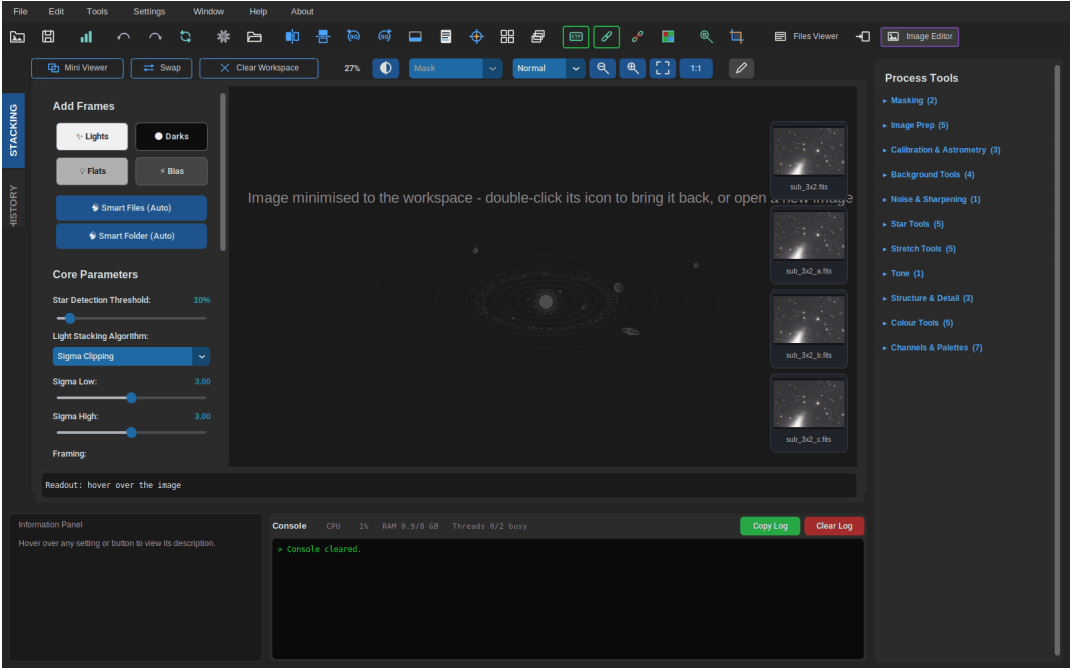


One image filling the editor, with the rest of the set waiting as icons down the right-hand edge. Its controls are in the top right corner.

## Icons

Minimising an image parks it as an icon carrying a thumbnail of itself. Icons cascade down the right-hand edge of the workspace, clear of the window controls. Double-click one to bring that image back; right-click it to rename or close it.

An icon you drag somewhere stays there. That is taken as a decision rather than a nudge: for the rest of the session that image returns to the spot you chose, through being opened, worked on and minimised again. Icons you have not moved carry on taking the next free slot in the cascade.

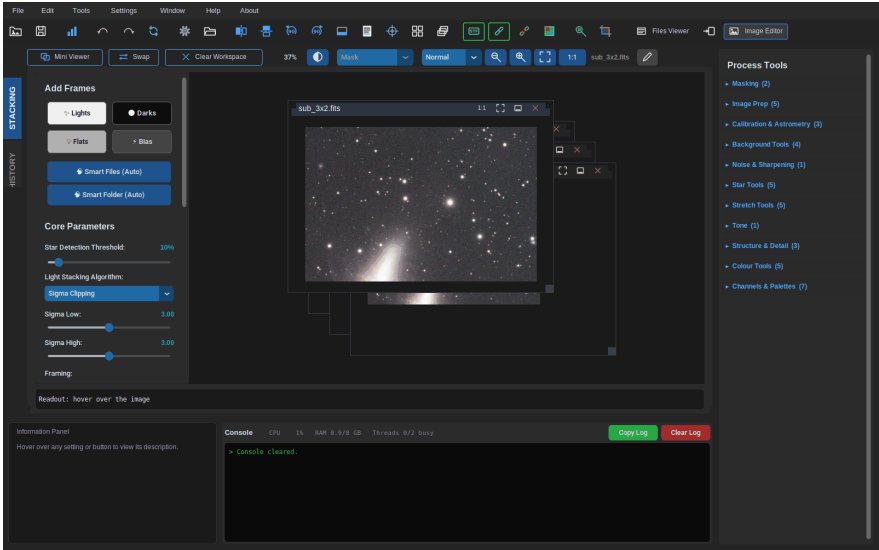


The same four frames, all minimised. The message in the middle says so, and names the way back.

### Arranging them all at once

The Window menu arranges everything on the workspace in one go, and both commands bring minimised images back up first - arrange ALL means all of them.

| Command         | What it does                                                                                                                                                                                                                     |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tile Windows    | Every image the same size, edge to edge. The grid is chosen for the shape of your frames rather than being fixed, so four landscape subs come out two by two and three come out in whichever arrangement shows the most picture. |
| Cascade Windows | Staggered, one title bar apart, so every name is readable. Useful for stepping through a set.                                                                                                                                    |
| Clear Workspace | Closes every open image. It always asks first, and the question names which images would lose unsaved work.                                                                                                                      |



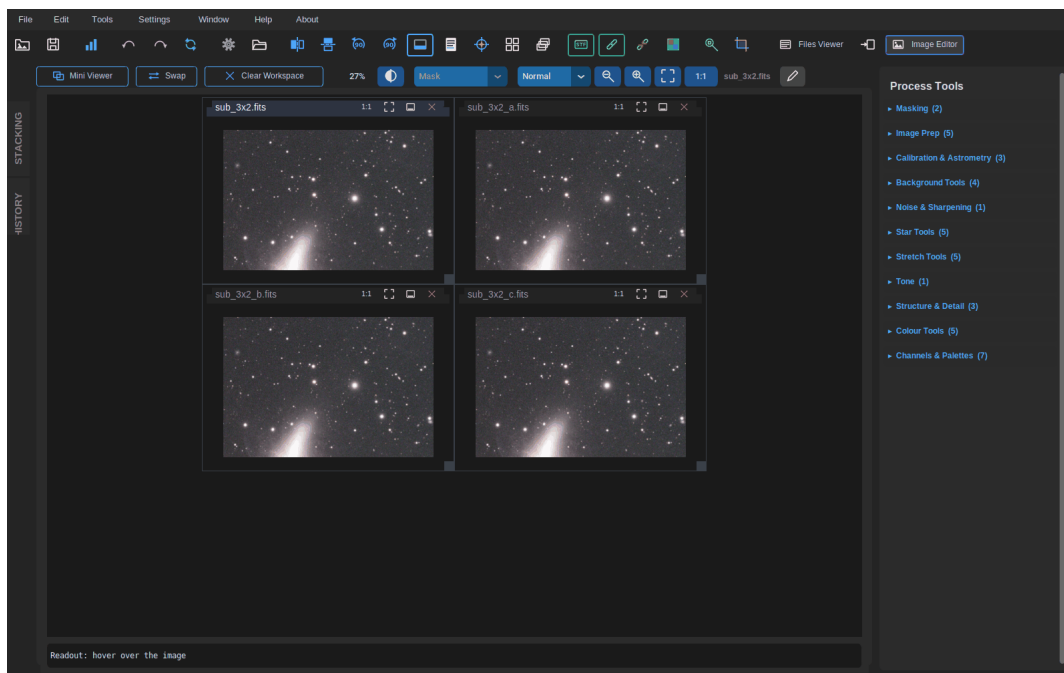
Cascade Windows.

The Window menu also lists every open image by name, minimised ones included - which is the quick way back to one when the workspace is busy.

## More room when you need it

Both side panels fold away. The button at the right-hand end of the toolbar hides the tools panel, and the console at the bottom has its own toggle. The images do not move when either one goes: the panel floats over the workspace rather than pushing it aside, so what you are looking at stays exactly where it is.

Tiled images keep the positions they were given, even after a panel folds away - the arrangement is worked out for the editor WITH both panels showing, precisely so that hiding one cannot shuffle your images around while you are looking at them. Tile again if you want them spread across the wider workspace.



Both panels folded away. The images have not moved.

## The editor toolbar

| Control         | What it does                                                                                                                                         |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mini Viewer     | Opens the current image in a separate small window with its own zoom, pan and Fit - useful for keeping an eye on one image while working on another. |
| Swap            | Exchanges which image the tools are pointed at.                                                                                                      |
| Clear Workspace | Closes everything on the workspace at once.                                                                                                          |
| Zoom %          | The current zoom, with buttons for zoom in, zoom out and fit to window. Double-clicking the picture returns it to fit.                               |
| Mask            | Which mask is active, if any, and the buttons to show, invert or delete it. See Chapter 6.                                                           |
| Blend           | How the active mask is applied.                                                                                                                      |

Below the picture, the readout line reports the pixel value under the pointer. Several tools use that same readout to let you pick a value off the image instead of guessing a number - the Pick buttons in RangeSelection, LinearFit and Background Neutralization all work that way.

## CHAPTER 5

## How editing works

Apply, undo, and what is actually being changed.

PhotonWorks edits the pixels, not a list of instructions. When a tool applies, the image in the Editor really has changed, and the next tool works on that result. This is the same model Siril and PixInsight use, and it is different from an adjustment-layer editor where everything stays reversible for ever.

### Preview, then apply

Most tools preview live in the Editor while you move their controls, and nothing is committed until you press the tool's own Run or Apply button. Closing a tool without applying leaves the image exactly as it was.

**Note.** A live preview is the real calculation on a downscaled copy, not an approximation of it - so what you see is what you get. Set Preview Region lets you keep one area at true full resolution while the rest stays on the fast preview.

### Undo and redo

Ctrl+Z steps back, Ctrl+Y steps forward, and both are also in the Edit menu and on the toolbar. The history is per image, so undoing in one window does not disturb another.

### Saving

Ctrl+S saves over the file you opened; Ctrl+Shift+S saves a copy somewhere else. The formats offered are:

|                     |                 |
|---------------------|-----------------|
| FITS Image          | .fits .fts .fit |
| XISF Image          | .xisf           |
| TIFF Image (16-bit) | .tif .tiff      |
| PNG Image           | .png            |
| JPEG Image          | .jpg .jpeg      |

**Note.** FITS and XISF keep the full 32-bit floating-point data and the header, including any plate solution. TIFF is 16-bit. PNG and JPEG are 8-bit and for sharing only - do not carry work forward in them.

**Careful.** Results that a tool writes to the scratch folder rather than to a file of your own - a starless image, split channels, a PixelMath result - are kept until the next launch clears them. Save anything you want to keep.

## CHAPTER 6

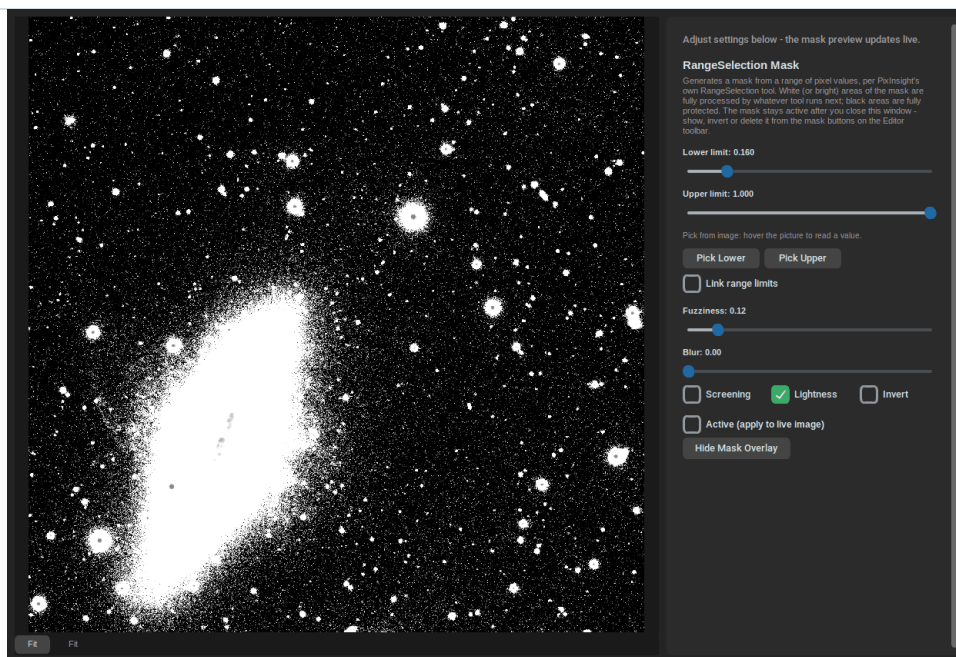
# Masks

Restricting a tool to part of the image.

A mask says where a tool is allowed to work. White areas are processed fully, black areas are left completely untouched, and grey is in between. Once a mask is active it applies to whatever tool you run next - every tool in the app, not just the one that made it.

There are two ways to make one, and they answer different questions.

|                     |                                                                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RangeSelection Mask | Selects by pixel VALUE. Give it a range of brightnesses and it selects everything in that range, wherever it is. Right for 'the bright parts', 'the stars', 'the sky'.                     |
| Shape Mask          | Selects by POSITION. Draw an ellipse or a polygon and that is the region. Right for 'this galaxy', which a value-based selection cannot separate from a background of the same brightness. |



RangeSelection Mask. The preview on the left is the mask itself: white is what the next tool will affect.

Both stay active after their window is closed, which is the point - you draw the region, close the window, then run the tool you wanted it for. The mask controls on the editor toolbar show, invert and delete the active mask, and the blend dropdown beside them controls how it is applied.

**Careful.** A mask you have forgotten about is the most common reason a tool appears to do nothing. If a tool has no visible effect, check the mask dropdown on the toolbar first.

## CHAPTER 7

## Tool reference

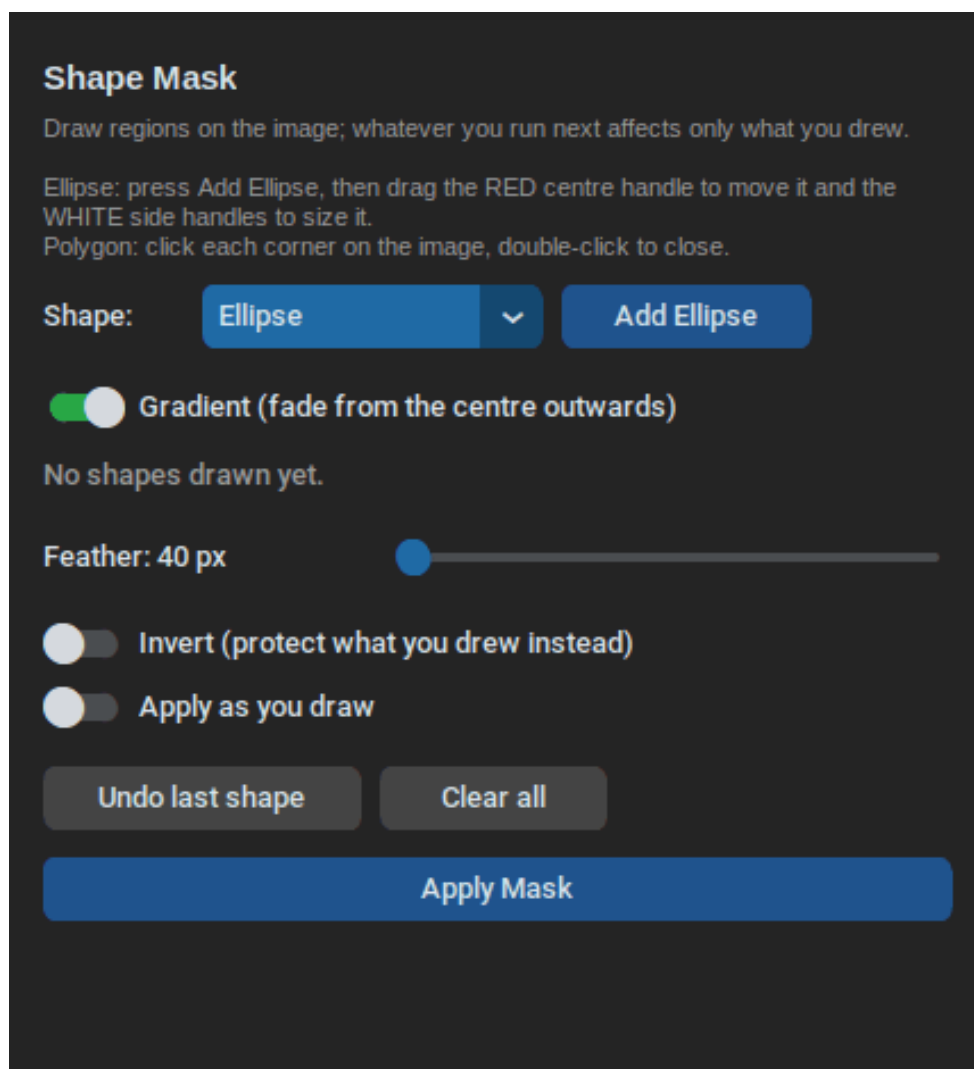
All 41 processing tools, in the 11 groups they appear in on the sidebar.

Each entry below is the tool's own description, as the app itself gives it when you hover over the button. The screenshots are of the real dialogs, on real data.

**Note.** The groups are collapsible drawers and all start closed. Opening one does not close the others, so several can be open at once. The Tools menu lists exactly the same tools in the same groups.

### Masking (2)

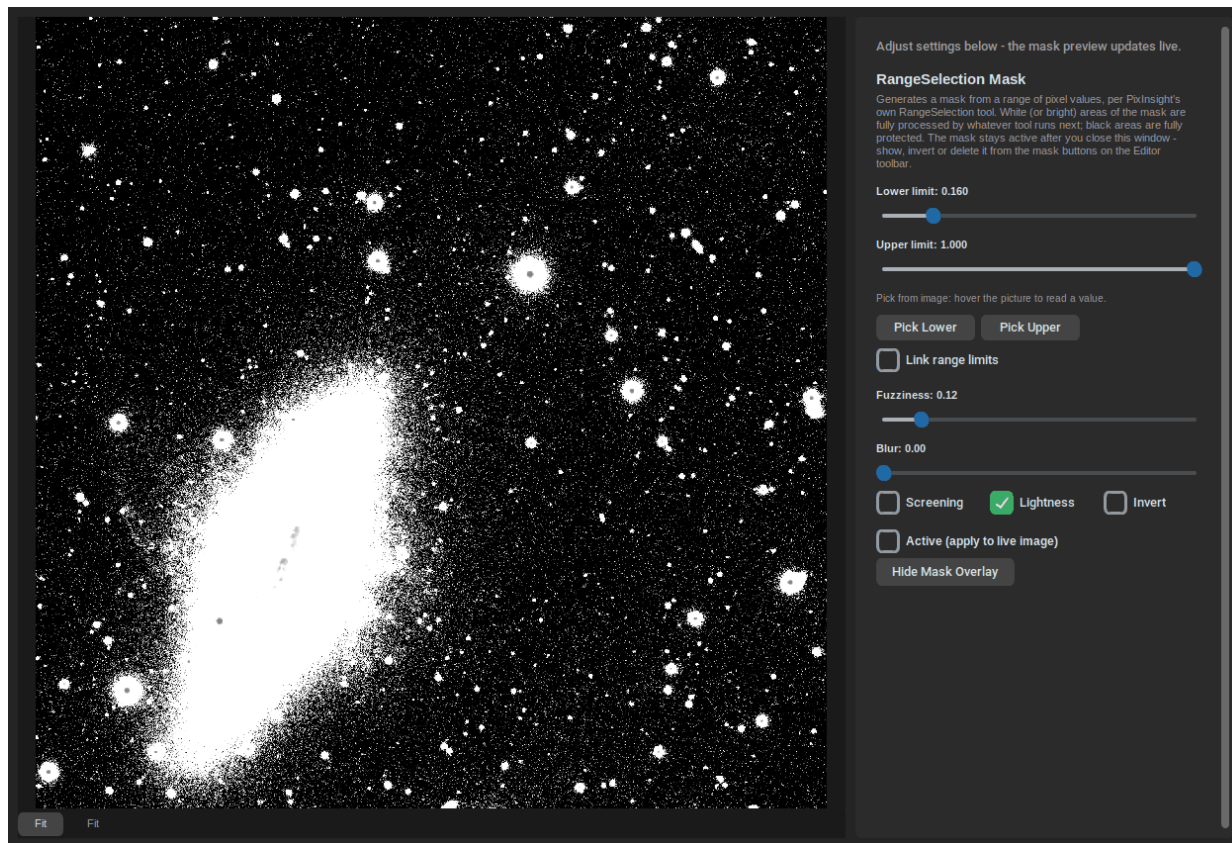
#### Shape Mask



Draw regions on the image - press Add Ellipse and drag its handles to size and place it, or build a polygon corner by corner - and use them as a mask. The companion to RangeSelection: that one selects by pixel VALUE, so it cannot separate a galaxy from a background of the same brightness. This selects by POSITION - you say where.

Shapes add together, the edge can be feathered so the transition does not show, and Invert protects what you drew instead of processing it. The mask stays active after you close the window, so draw the region then run whatever tool you wanted it for.

## RangeSelection Mask



Generates a mask from a range of pixel values, per PixInsight's own RangeSelection tool - Lower/Upper limit - pickable straight off the image, so the mask starts exactly at your own sky level rather than at a guessed number - Fuzziness (soft transitions), Blur (PixInsight calls it Smoothness), Screening (proportional rather than flat white), Lightness (CIE L\* vs per-channel RGB/K), Invert. Check 'Active' to apply it to the live image: white areas get fully processed by whatever tool runs next, black areas stay completely untouched - works with any tool in the app, not just this one. The mask stays active after you close this window - use the mask buttons on the Editor toolbar to show, invert or delete it.

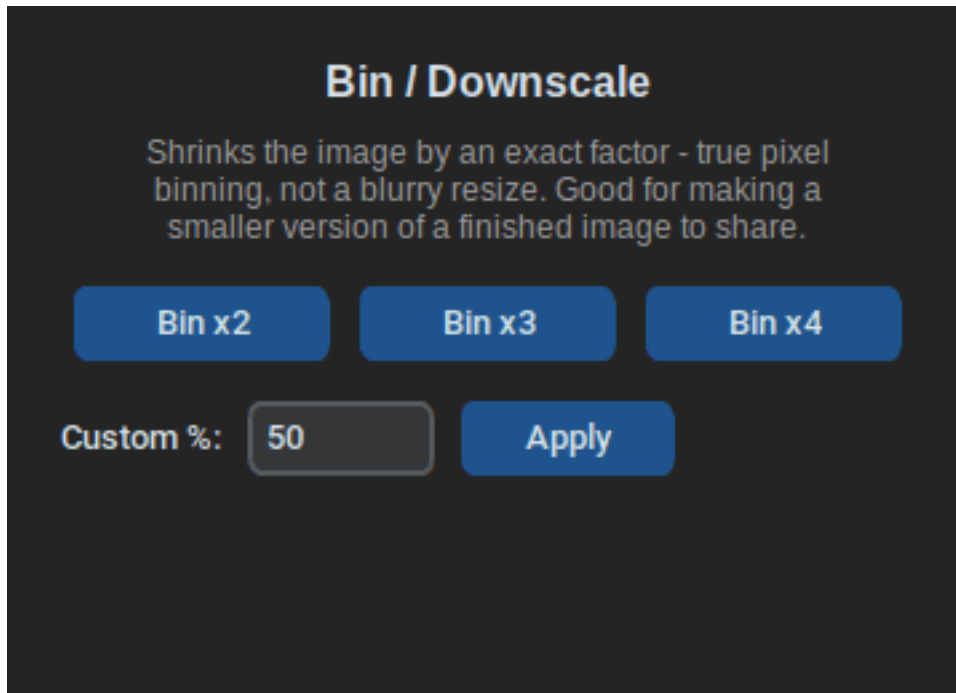
## Image Prep (5)

## View FITS Header

```
SIMPLE      = True
BITPIX      = -32
NAXIS       = 3
NAXIS1      = 1150
NAXIS2      = 1150
NAXIS3      = 3
OBJECT      = M82
IMAGETYP    = LIGHT
EXPTIME     = 300.0
STACKCNT    = 84
LIVETIME    = 25200.0
FILTER      = LRGB
INSTRUME    = ASI2600MM Pro
TELESCOP    = SW Esprit 100ED
FOCALLEN    = 550.0
XPIXSZ      = 3.76
YPIXSZ      = 3.76
CCD-TEMP    = -10.0
GAIN        = 100
OFFSET      = 50
XBINNING    = 1
YBINNING    = 1
DATE-OBS    = 2026-03-14T21:42:11
SITELAT     = 53.4084
SITELONG    = -2.9916
BAYERPAT    = NONE
PROGRAM     = PhotonWorks
STACKMTH    = Winsorized Sigma
```

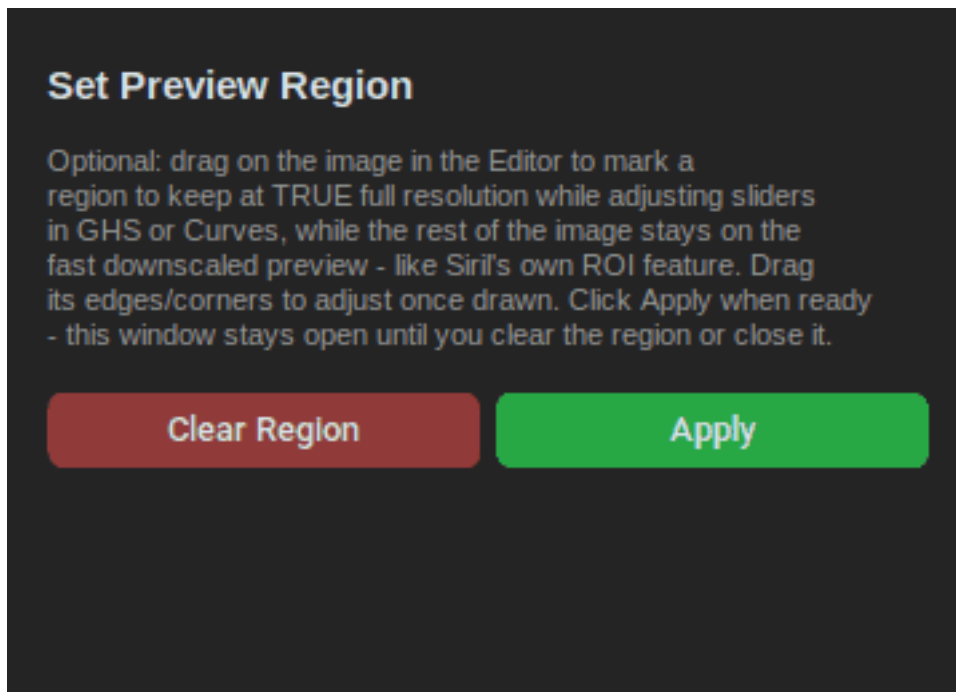
Shows the raw FITS header (exposure, temperature, filter, WCS, etc.) for the file currently being previewed.

## Bin/Downscale



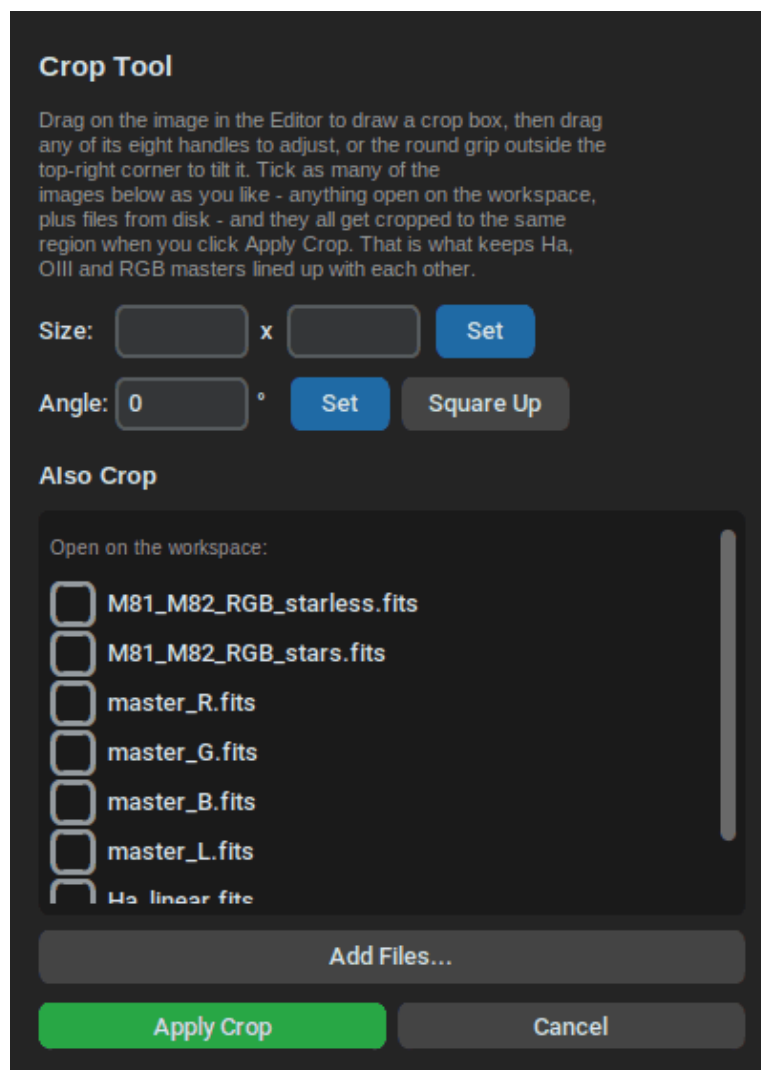
Shrinks the image by an exact integer factor (true pixel binning - averages each block down to one pixel, not a blurry resize) or a custom percentage. Useful for making a smaller version of a finished image to share online. Applies directly, same as GHS/Curves - use Undo to revert.

## Set Preview Region



Optional: mark a region to keep at TRUE full resolution while adjusting sliders in GHS or Curves, while the rest of the image stays on the fast downscaled preview - like Siril's own ROI feature. Drag out a region, then adjust it with any of its eight handles - four corners and four sides - or drag the middle to move it. Off by default and doesn't replace the normal preview.

## Crop Tool

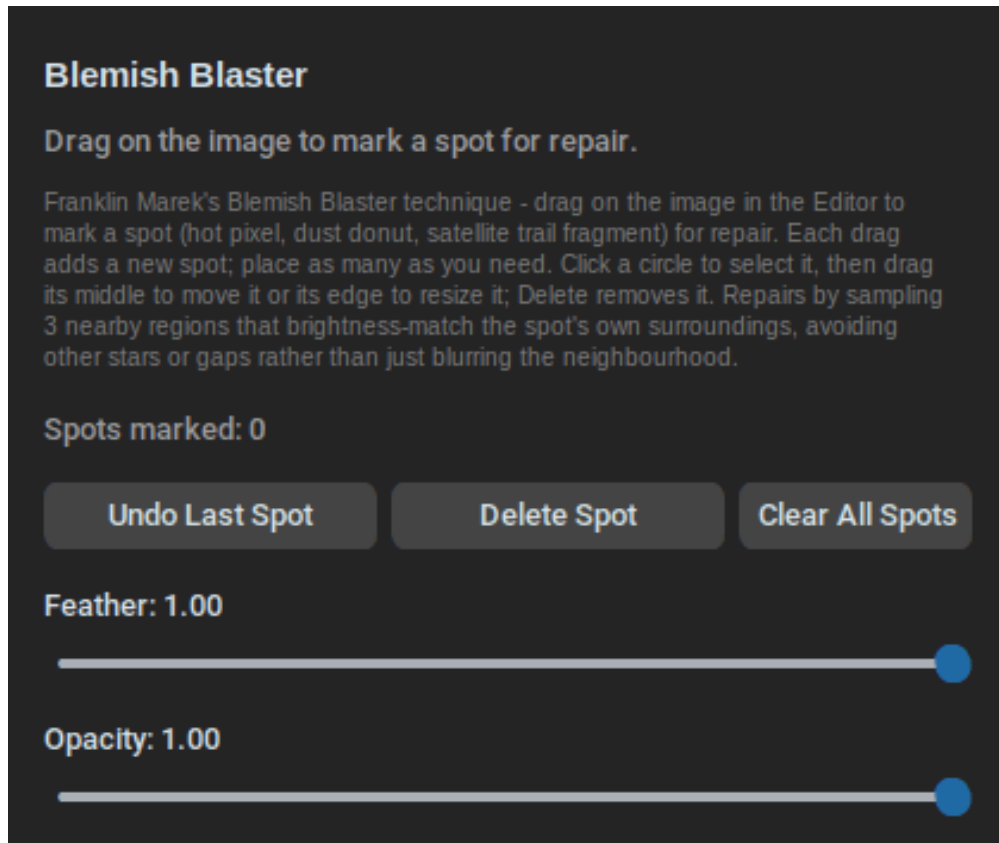


Draw a crop box on the image, then drag any of its eight handles - four corners and four sides - to adjust, or type an exact size, and click Apply Crop when ready.

The round grip just outside the top-right corner TILTS the box: drag it and the box turns, and a tilted crop comes out upright at the box's own size. That is how you square up a mosaic panel without rotating the whole frame first. Square Up puts it back to zero, where the crop stays an exact pixel-for-pixel cut with nothing resampled.

Also Crop lists every other image open on the workspace, so you can tick as many as you like and cut them all to the same region - which is how you keep two masters on the same pixel grid. Add Files... takes images from disk that are not open at all. Mini Viewers are still offered below those, as the legacy route.

## Blemish Blaster

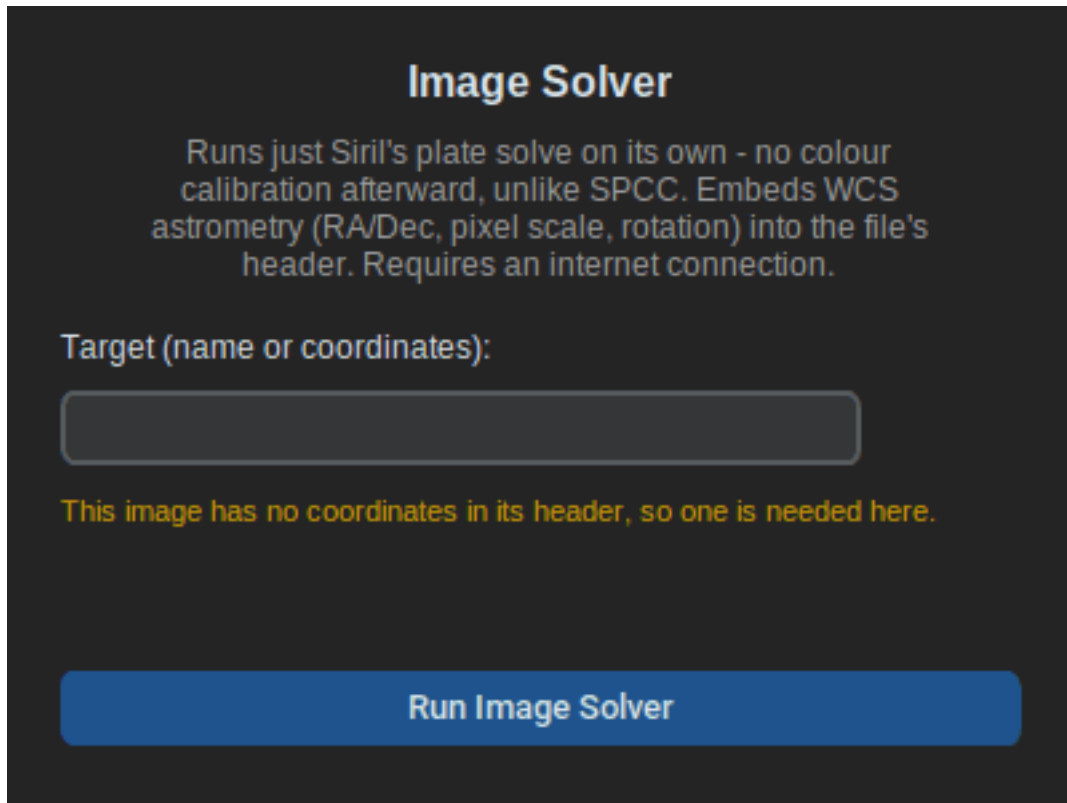


Franklin Marek's Blemish Blaster technique - drag on the image to mark a spot (hot pixel, dust donut, satellite trail fragment) for repair. Spots stay editable: click a circle to select it, drag its middle to move it or its edge to resize it, and Delete Spot (or the Delete key) removes just that one. Repairs by sampling 3 nearby regions that brightness-match the spot's own surroundings, avoiding other stars or gaps rather than just blurring the neighbourhood. Feather and Opacity sliders control how strong and how soft the repair is.

## Calibration & Astrometry (3)

---

## Image Solver



Runs just Siril's plate solve on its own, embedding WCS astrometry (RA/Dec, pixel scale, rotation) into the file's header - no colour calibration afterward, unlike SPCC below. On a FITS file it writes the solution straight back, since only header metadata changes; XISF and TIFF are deliberately left untouched and the console says so. Requires an internet connection - run this after cropping to avoid losing the plate solve to the crop.

## SPCC

### Plate Solve + Spectrophotometric Color Calibration

M81\_M82\_RGB.fits

Refresh Lists from Siril

OSC Sensor

Click Refresh Lists ->

White Reference

Average Spiral Galaxy

☐ Narrowband / Dual-Band Filter (e.g. L-eXtreme, L-Enhance, Duo-Band)

OSC Filter

Click Refresh Lists ->

Target (name or coordinates):

This image has no coordinates in its header, so one is needed here.

Run Plate Solve + SPCC

Runs Siril's plate solve + Spectrophotometric Color Calibration on the currently previewed image. Requires an internet connection. Run this after cropping to avoid losing the plate solve to the crop.

## LinearFit

### LinearFit

Select a reference image below to begin.

Fits a robust straight line matching this image's pixel values to a reference image's, per channel, then applies that line to the whole image. Useful for matching brightness/background levels across separately-processed masters (e.g. narrowband filters) before combining them. Images must be pixel-for-pixel registered (same size, same alignment) for the result to mean anything.

**Target Image (currently open)**

M81\_M82\_RGB.fits

**Reference Image**

(no reference image chosen) ▾

**Reject Low: 0.000**

●

**Reject High: 0.920**

●

Pick either end of the fit range from the image rather than guessing a number.

Pick Reject Low      Pick Reject High

Run LinearFit

Fits a robust straight line matching the currently-open image's pixel values to a separate reference image's, per channel, then applies that line to the whole image. Useful for matching brightness/background levels across separately-processed masters (e.g. different narrowband filters) before combining them. Images must be pixel-for-pixel registered.

Pick Reject Low and Pick Reject High set the ends of the fit range by clicking the image - click what you want kept, not what you want out.

One dropdown chooses the reference: the images open on the workspace, then any Mini Viewers, then Browse for a file. Choosing one loads it straight away.

## Background Tools (4)

---

## Manual Background Extraction

### Manual Background Extraction

Samples are placed on SKY and a surface is fitted through them, then removed. Click the image to add a sample, drag to move one, right-click to delete. Each new sample slides downhill onto the dimmest spot near where you drop it, so one landing on a star corrects itself.

Once there are 20 or more, each sample is coloured by how it compares with the others, channel by channel: GREEN reads like sky, AMBER stands out, RED is measuring something else. Move or delete the red ones. Below 20 they stay green - with a few points there is nothing to compare against.

Grid:  8 x 8

0 sample(s).

Fit

Degree:  2

☒ Refine with RBF  0.10

Mode:

☐ Show the background model

Place samples on sky, then Apply.

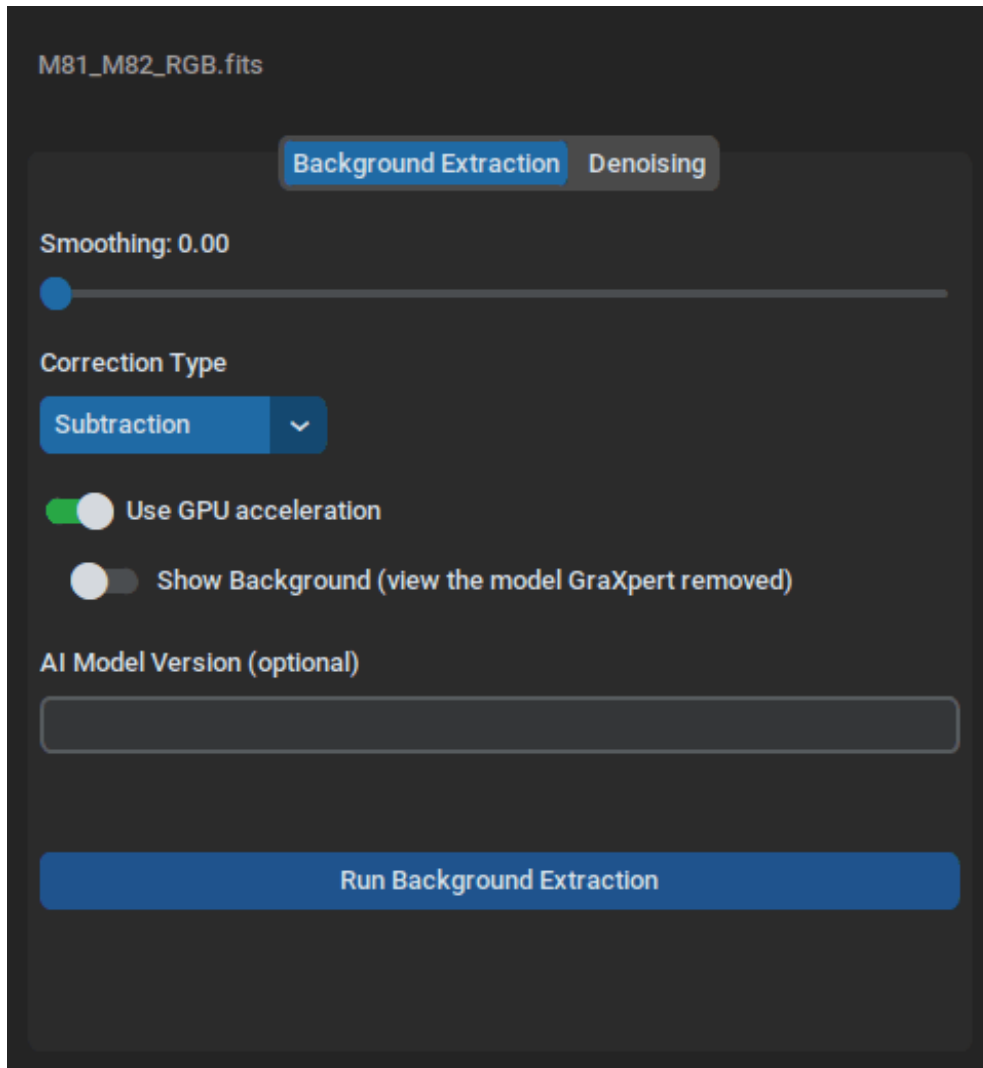
Places sample points on the SKY and fits a background surface through them, then removes it - the manual counterpart to the two automatic tools below.

The automatic ones choose their own sample positions, which is right on a clean gradient and wrong the moment the target fills the frame: they fit the nebula and then subtract it. Here you decide where the sky is.

Samples slide downhill onto the dimmest spot near where you drop them, so one landing on a star corrects itself.

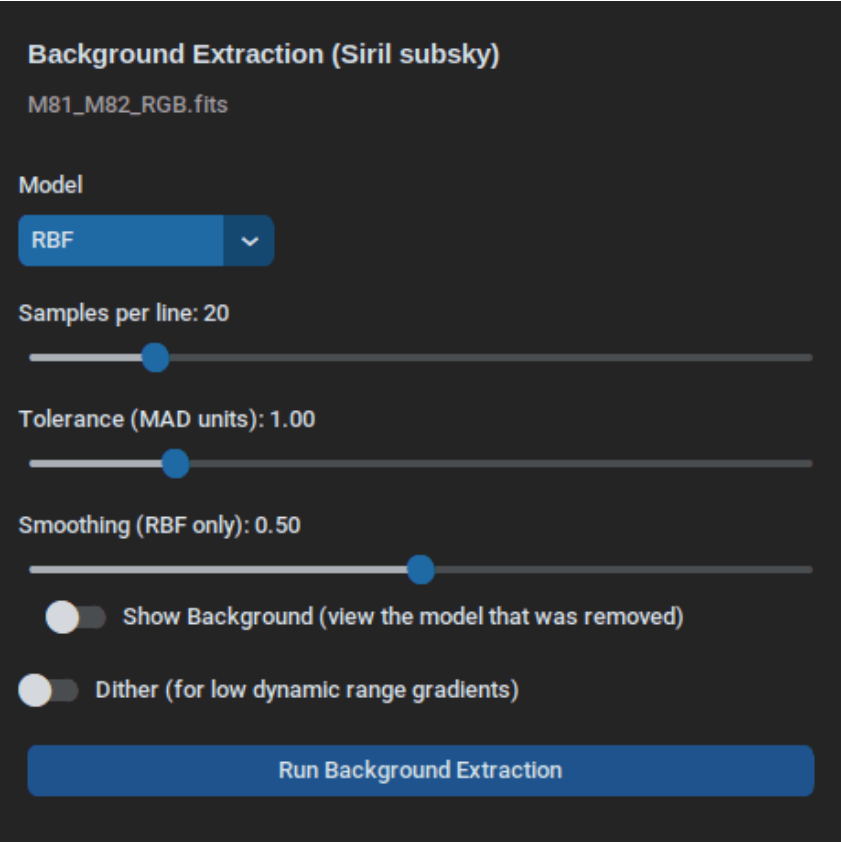
Click to add a sample, drag to move it, right-click to remove it. Exclusion areas work the same way: click corners to draw one, right-click to close it, Backspace to take back the last corner, and right-click inside a finished area to remove just that one.

## GraXpert



Runs GraXpert's AI Background Extraction or Denoising on the currently previewed image. Requires the GraXpert path to be set in Advanced Settings -> AI Tools.

## Background Extraction



**Background Extraction (Siril subsky)**

M81\_M82\_RGB.fits

Model

RBF

Samples per line: 20

Tolerance (MAD units): 1.00

Smoothing (RBF only): 0.50

☐ Show Background (view the model that was removed)

☐ Dither (for low dynamic range gradients)

Run Background Extraction

Runs Siril's background extraction on the currently previewed image - RBF, or Polynomial with a Degree you choose. Show Background displays the model it built rather than the corrected image, which is the way to tell a real gradient from real nebulosity it is about to remove. Useful for applying it AFTER other processing like NoiseXTerminator instead of only automatically before stacking, since running noise reduction on already-extracted data can give patchy results.

## Background Neutralization

Adjust settings below - the preview updates live.

Equalizes the average red, green and blue background levels to a neutral grey, removing any colour cast in the sky background. Draw a Preview Region over open sky in the Image Editor first (optional) to sample just that area; otherwise the whole image is used as the background reference.

No Preview Region set - using the whole image as the background reference.

Lower Limit: 0.000

Upper Limit: 0.100

Pick Background sets both limits from a patch of your own sky.

Pick Background

Working Mode

Rescale as needed

Run Background Neutralization

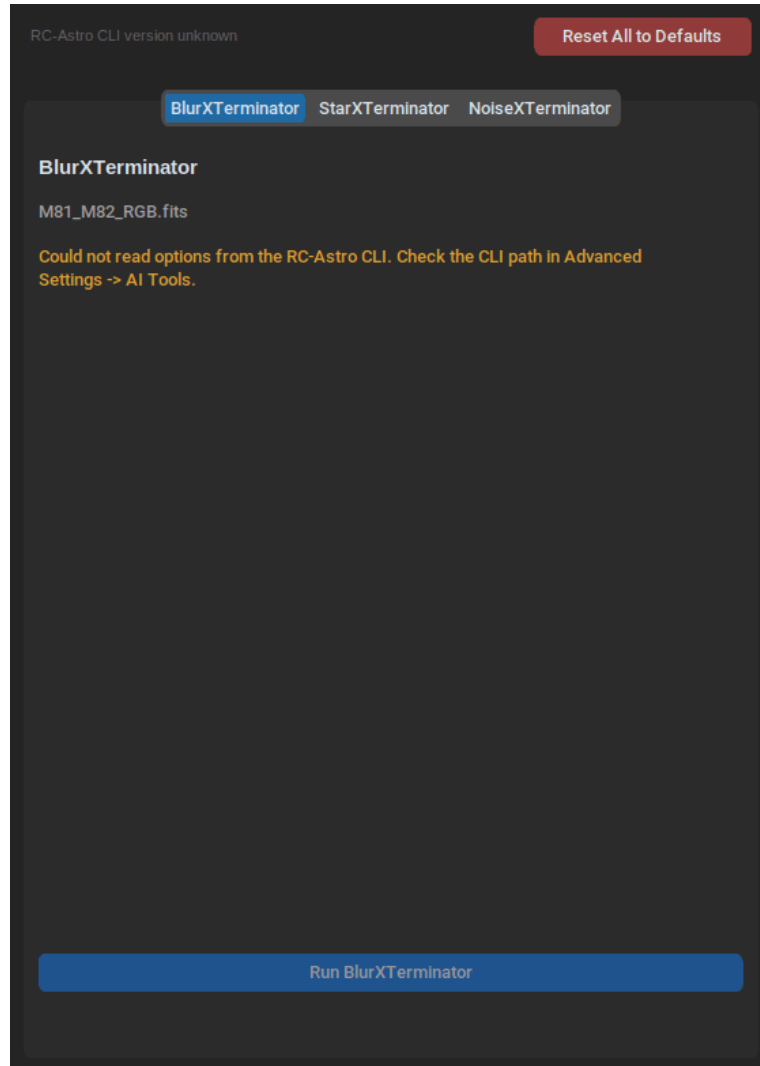
Equalizes the average red, green and blue background levels to a neutral grey, removing any colour cast in the sky background. A neutral background is a precondition for correct colour calibration, so this normally runs before SPCC or other colour calibration steps. Draw a Preview Region over open sky first to sample just that area, or leave it unset to use the whole image.

Pick Background sets both limits by clicking a patch of your own sky, rather than guessing the numbers.

## Noise & Sharpening (1)

---

## Xterminator Tools



Opens BlurXTerminator, StarXTerminator, and NoiseXTerminator in one tabbed window, each running on the currently previewed image via the RC-Astro CLI. Requires the CLI path to be set in Advanced Settings -> AI Tools.

## Star Tools (5)

---

## RGB Stars to NB

### RGB Stars to Narrowband

Pick the three images below.

Puts REAL star colour, measured from broadband data, onto a narrowband image. Different from NB to RGB Stars, which SYNTHESISES star colour when you have no broadband data at all - use this one when you DO.

Needs three images, all registered to each other and already stretched:

- your broadband RGB frame WITH stars (an unfiltered OSC frame counts)
- the same RGB frame with the stars removed
- your narrowband image with the stars removed

Ported from Bill Blanshan's RGB Stars-to-Narrowband Transfer V14.

RGB with stars

(no image chosen)

▼

RGB starless

(no image chosen)

▼

Narrowband starless

(no image chosen)

▼

Stretch Control: 0.050

Star Mask Strength: 0.50

Amount: 1.00

Reset Sliders

Run Transfer

Puts REAL star colour, measured from BROADBAND data, onto a narrowband image - the complement of NB to RGB Stars, which synthesises star colour when you have no broadband data at all. Needs three registered, stretched images: your RGB frame with stars (an unfiltered OSC frame counts, but NOT one shot through a narrowband filter - its star colour is already false), the same RGB frame starless, and your narrowband image starless. It recovers the star layer from the RGB by unscreening, screens it onto the narrowband, then does a CIELAB transfer taking lightness from one and colour from the other, which is what makes it more than a screen blend. Away from stars the narrowband is very closely preserved, though not bit-exactly - it passes through a clamp, a global rescale and a Lab round trip on the way. Ported from Bill Blanshan's RGB Stars-to-Narrowband Transfer V14.

## Star Reduction

### Star Reduction

Bill Blanshan's Star Reduction PixelMath methods

Select both images below to begin.

Needs a starless version of your image (e.g. from StarXTerminator via Xterminator Tools) alongside the original - reduces stars by recombining the two, rather than trying to detect and shrink stars directly, which is what makes this reliable regardless of star size.

Original Image (with stars)

Browse...

Starless Image

Browse...

Method

Transfer

▼

Run Star Reduction

Reduction Strength (S): 0.15

Bill Blanshan's Star Reduction PixelMath methods (Transfer/Halo/Star) - needs a starless version of your image (e.g. from StarXTerminator) alongside the original. Select both images to begin.

## Screen Stars

### Screen Stars

Bill Blanshan & Mike Cranfield's ScreenStars technique

Select both images below to begin.

Recombines a starless image with a separately-processed stars-only image via the Screen blend - the exact inverse of the "unscreen" math StarXTerminator/StarNet use to remove stars, so this reconstructs them cleanly rather than just adding brightness on top.

Starless Image

M81\_M82\_RGB.fits (in Editor) ▼

Stars-Only Image

(no image chosen) ▼

Run Screen Stars

Recombines a starless image with a separately-processed stars-only image via the Screen blend - Bill Blanshan & Mike Cranfield's technique, the exact inverse of the "unscreen" math used to remove stars in the first place. Select both images to begin.

## Star Stretch

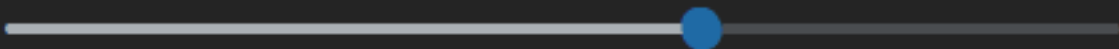
Adjust settings below - the preview updates live.

### Star Stretch

Franklin Marek's Star Stretch v2.1 technique

A linear-to-nonlinear stretch tuned for a stars-only image - meant to run on a combined stars image before screening it back into a starless one. Boosts colour and removes any green cast automatically on colour images (skipped for mono).

Stretch Amount: 5.00



Color Boost Amount: 1.00



Franklin Marek's Star Stretch v2.1 technique - a linear-to-nonlinear stretch tuned for a stars-only image, boosting colour and removing green cast automatically. Meant to run on a combined stars image before screening it back into a starless one.

## NB to RGB Stars

### NB to RGB Stars

Select images below to begin.

Franklin Marek's NBtoRGBStars v1.6 technique - combines Ha/OIII/ (optional SII) stars-only images into a realistic RGB stars image, ready to screen back into a starless one. Select either the 3 separate mono images, OR a single OSC image shot through a dual-band (Ha+OIII) filter - if an OSC image is selected, it takes priority over the mono pickers.

Ha Stars Image

(no image chosen) ▾

OIII Stars Image

(no image chosen) ▾

SII Stars Image (optional)

(no image chosen) ▾

OR

OSC (for dual band filter) Image

(no image chosen) ▾

☐ Ha to OIII ratio (Optional)

☐ Apply Star Stretch (Recommended)

Franklin Marek's NBtoRGBStars v1.6 technique - combines Ha/OIII/(optional SII) stars-only images into a realistic RGB stars image, ready to screen back in. Also accepts a single OSC image shot through a dual-band filter instead of 3 separate mono ones. Optional Apply Star Stretch runs that technique on the result as the final step.

## Stretch Tools (5)

---

## Simple Stretch

**Adjust settings below - the preview updates live.**

Permanently applies an auto-stretch to the actual pixel data - the "STF Stretch" toggle only ever affects what you see, not what's saved. Different images need different amounts, so preview updates live as you adjust Stretch Amount below; needed before tools like Narrowband Normalization if your image is still linear/unstretched.

**Stretch Mode**

☒ **Linked (preserves colour balance)**

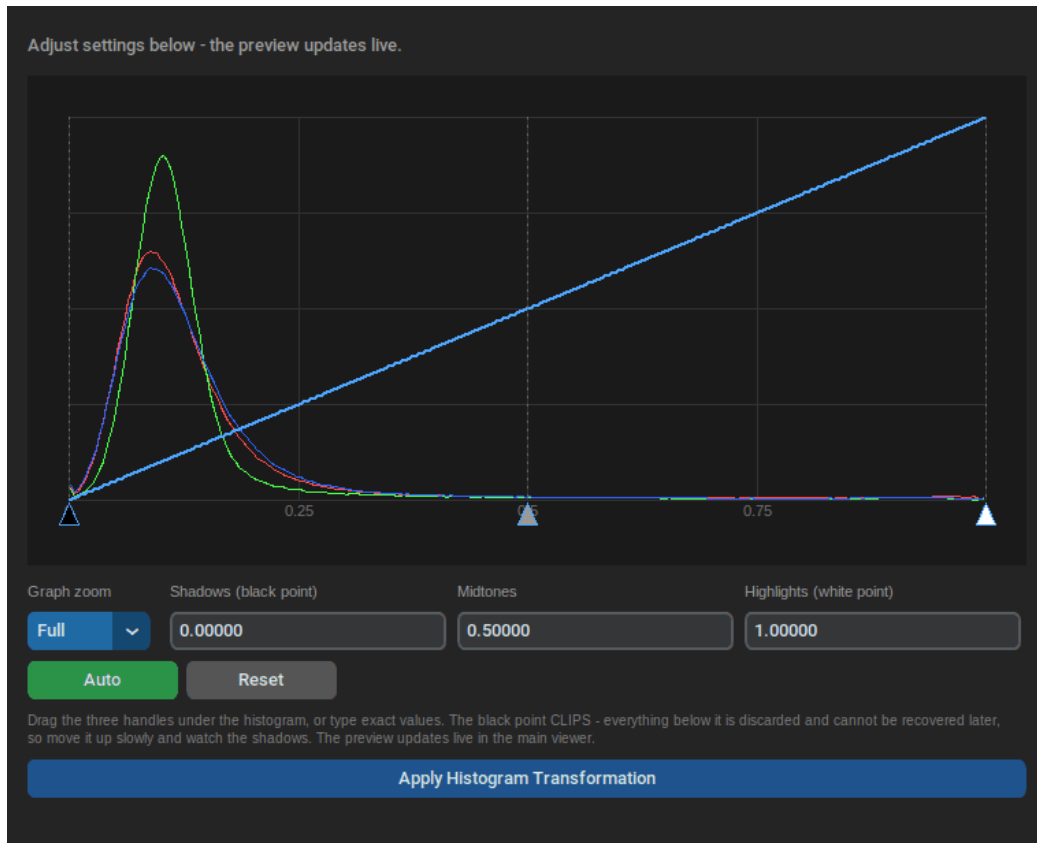
☐ **Unlinked (stretches each channel independently)**

**Stretch Amount: 0.25 (default)**

**Run Simple Stretch**

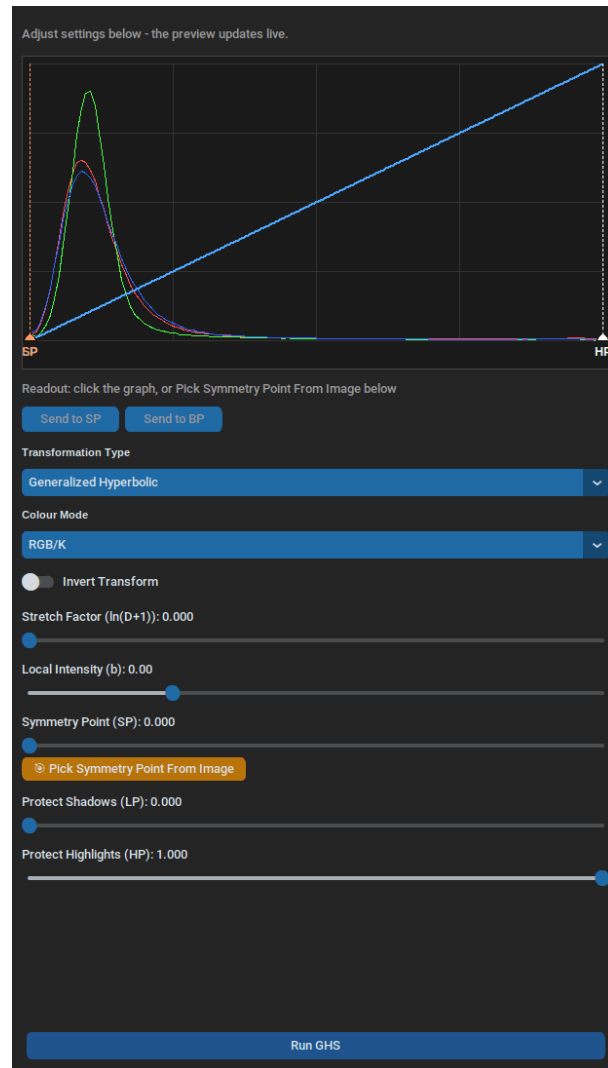
Permanently applies the same auto-stretch algorithm used for the viewer's own display to the actual pixel data - a quick one-slider stretch. Needed before tools that expect a stretched image (e.g. Narrowband Normalization) if you haven't stretched elsewhere.

## Histogram Transformation



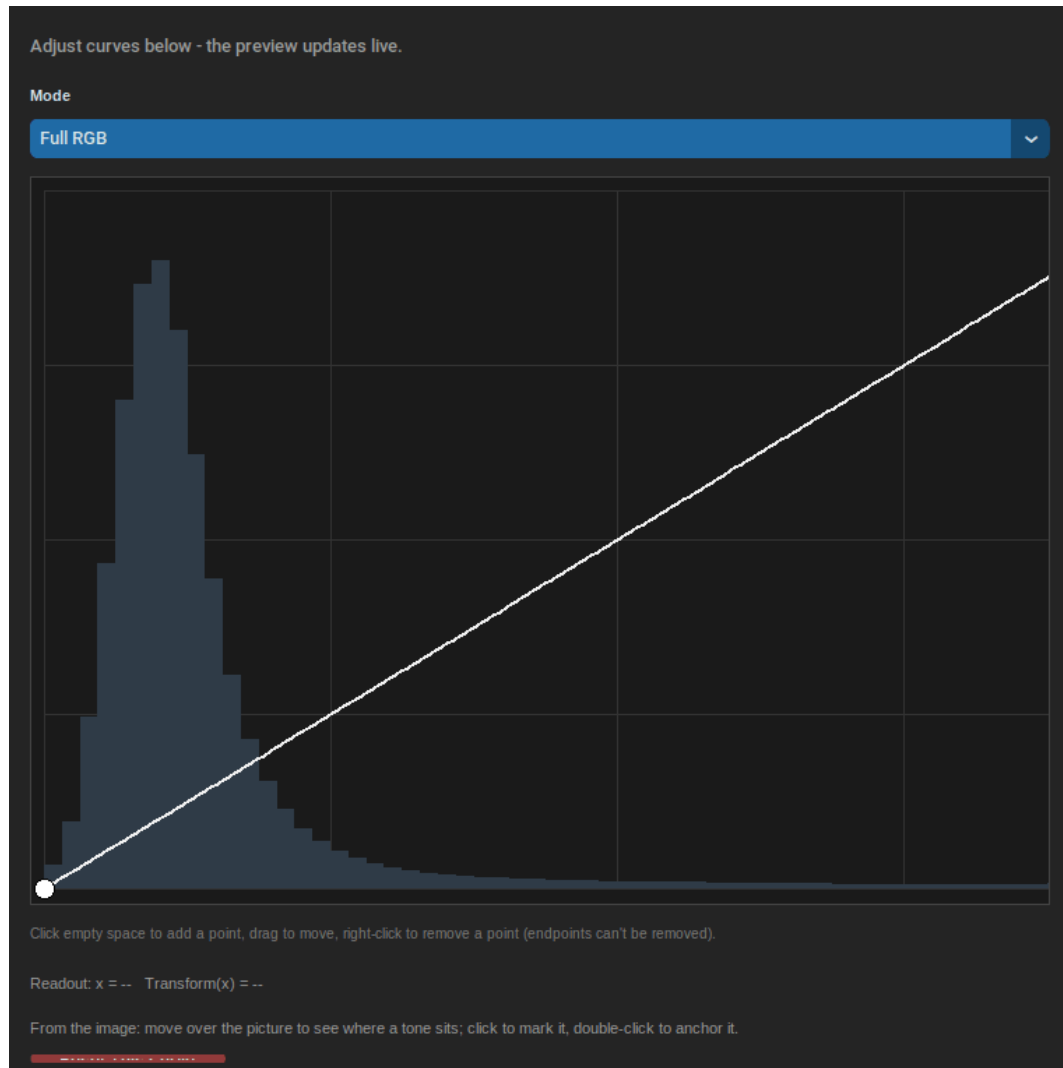
PixInsight-style HistogramTransformation - drag the black, midtone and white handles under a live RGB histogram, with the transfer curve drawn against the identity diagonal. Auto suggests a starting black point and midtone from the image itself. Linked across channels. Live preview in the Editor. Note the black point CLIPS: everything below it is discarded permanently.

## GHS



Built from scratch in this app (not scripted through Siril) from the published GHS specification (David Payne & Mike Cranfield) - Generalized Hyperbolic, Midtone Transfer, Arcsinh, Power Law, or Linear, with full control over strength, symmetry, and shadow/highlight protection, plus Colour/Lightness/Saturation modes. Live preview.

## Curves



Interactive curve editor - Full RGB, Individual RGB, Luminance Only, or Saturation. Click to add a point, drag to move it, right-click to remove it. Built directly in this app (not scripted through Siril).

## VeraLux HyperMetric Stretch

Adjust settings below - the preview updates live.

### VeraLux HyperMetric Stretch

by Riccardo Paterniti - ported from the real Siril script

Linear-to-nonlinear stretch - run before other stretch tools, same as Simple Stretch/GHS. Stretches one luminance channel, then reconstructs colour via each pixel's own channel-to-luminance ratio rather than stretching R/G/B independently, avoiding the hue shift a plain stretch can cause.

**Processing Mode**

☒ Ready-to-Use (Aesthetic)  
☐ Scientific (Preserve)

**Sensor Profile**

Rec. 709 (Recommended) ▾

Target Background: 0.20

☒ Adaptive Anchor

< Auto-Calculate Log D

Log D (Intensity): 2.00

Protect b (Highlights): 6.00

Star Core Recovery: 3.50

**Color Strategy**

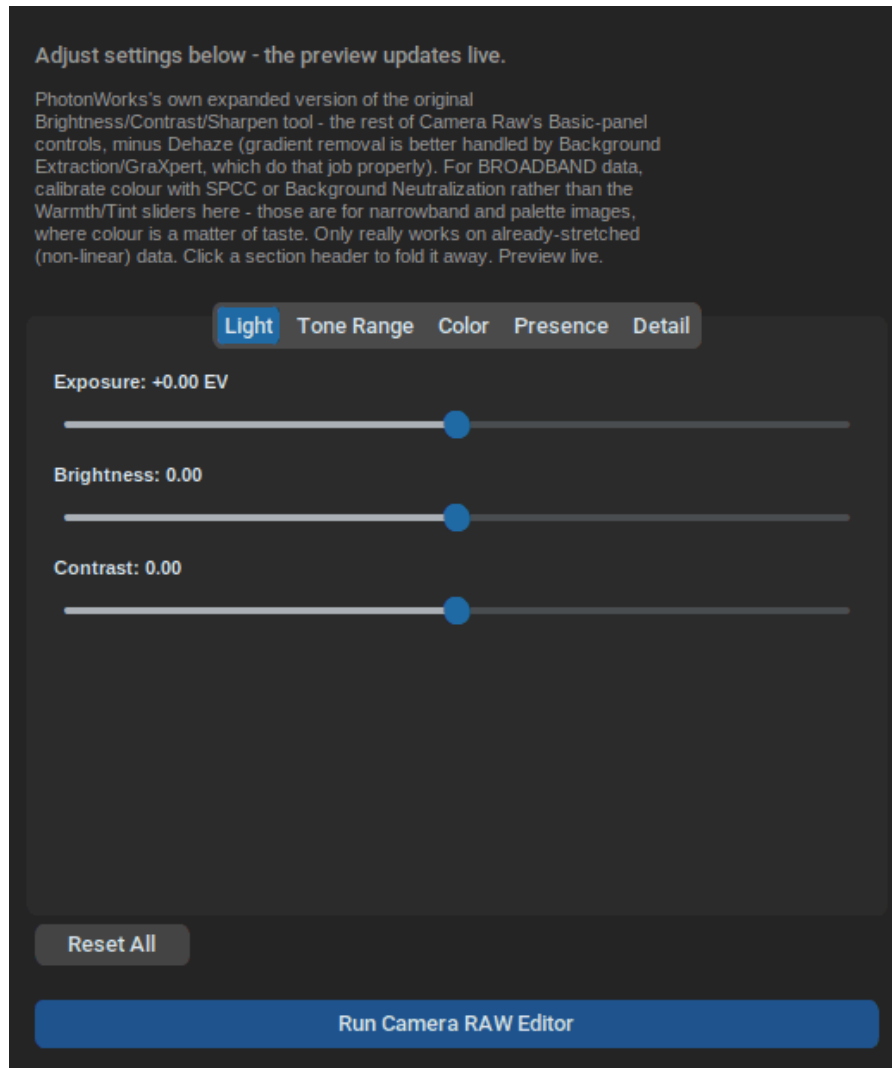
Balanced (Pure Vector)

Run VeraLux Stretch

A ported, verified version of Riccardo Paterniti's real VeraLux HyperMetric Stretch script for Siril - a linear-to-nonlinear stretch that preserves each pixel's original colour ratios rather than stretching R/G/B independently, avoiding the hue shift a plain histogram stretch can cause. Ready-to-Use (finished, export-ready look) or Scientific (raw, deterministic - a base for further manual tone mapping) modes, with sensor-specific luminance weighting and an Auto-Calculate solver.

## Tone (1)

## Camera RAW Editor

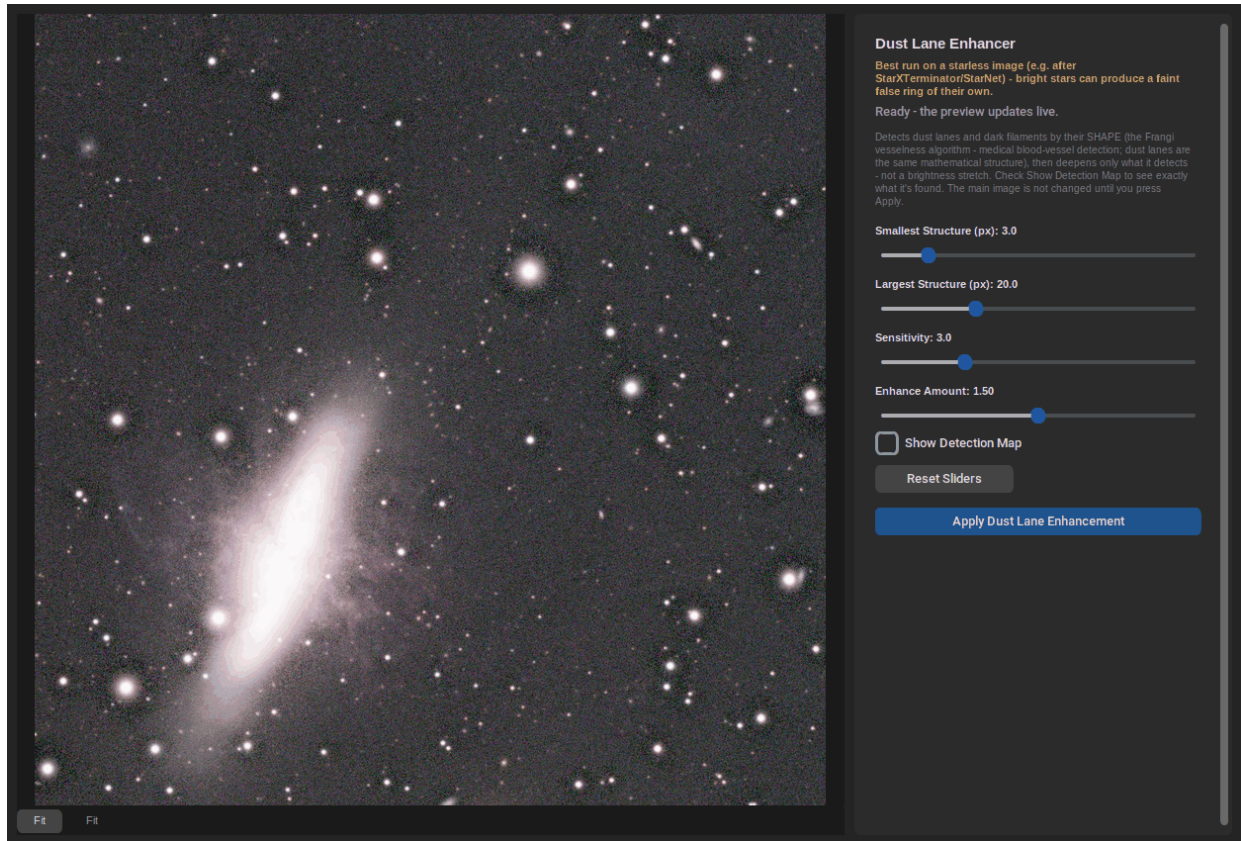


PhotonWorks's own expanded version of the original Brightness/Contrast/Sharpen tool - adds Exposure, Highlights/Shadows/Whites/Blacks, Vibrance, Clarity/Texture, Warmth, Tint, Hue Rotation and an adjustable-radius, noise-aware Sharpen, alongside the original Brightness/Contrast/Saturation/Sharpen. Live preview. Only really works on already-stretched (non-linear) data - on linear data, where almost all the signal sits near black, most of these controls behave unpredictably or barely at all.

## Structure & Detail (3)

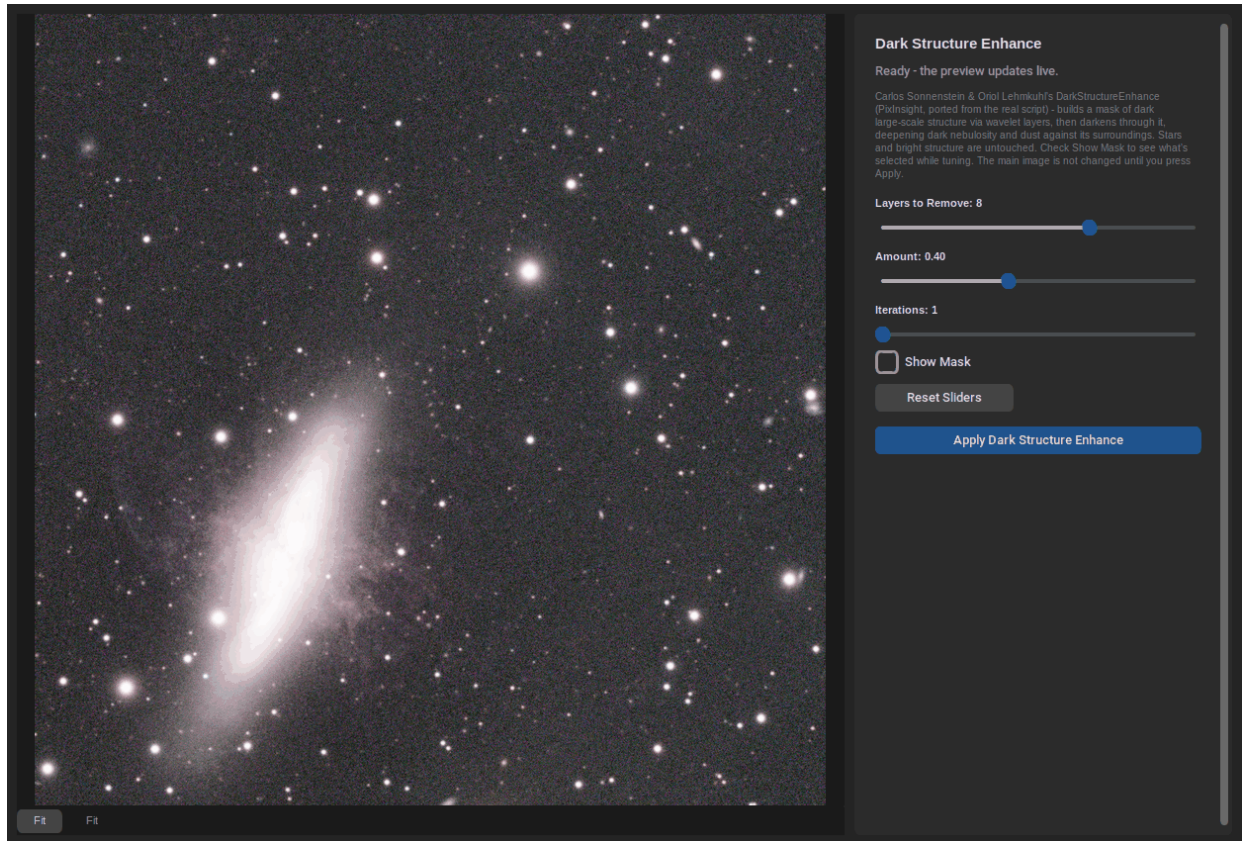
---

## Dust Lane Enhancer



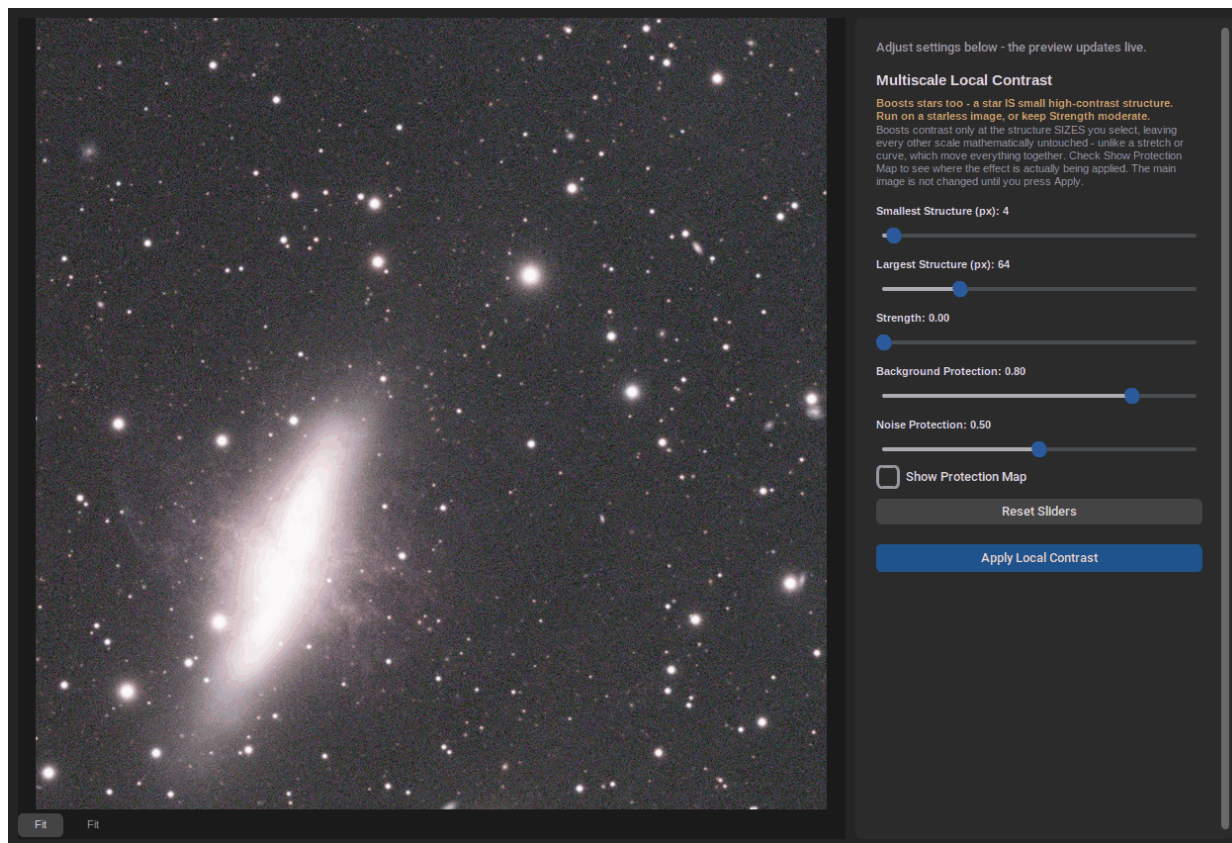
Detects dust lanes and dark filaments by their SHAPE - the Frangi vesselness algorithm from medical blood-vessel detection (dust lanes are the same mathematical structure: elongated ridges) - then deepens only what it detects. Not a brightness stretch: a shape-based enhancement no other astro processing software has. Works best on a starless image - bright stars can produce a faint ring-like false response of their own. Use Show Detection Map to see exactly what it's found while tuning.

## Dark Structure Enhance



Carlos Sonnenstein & Oriol Lehmkuhl's DarkStructureEnhance (the real PixInsight script, ported faithfully) - builds a mask of dark large-scale structure via wavelet layer decomposition, then darkens through it, deepening dark nebulosity and dust against its surroundings while stars and bright structure stay untouched. Companion to the Dust Lane Enhancer: this boosts dark structure generally, whatever its shape, vs that tool's filament-specific detection. Layers controls what counts as large-scale; Show Mask displays exactly what's selected.

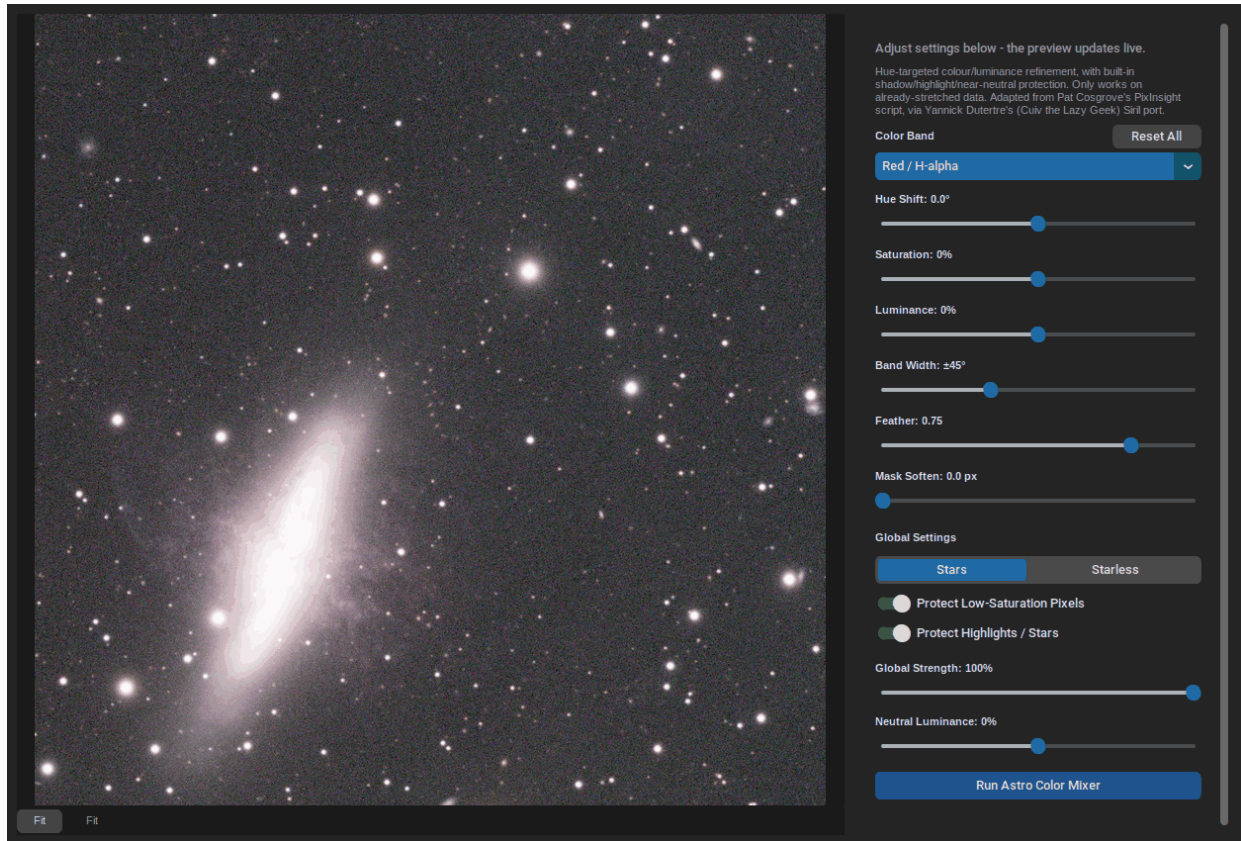
## Multiscale Local Contrast



Boosts contrast only at the structure SIZES you choose, leaving every other scale mathematically untouched - a genuinely different lever from a stretch or curve, which move every scale together. Splits the image into wavelet scale layers (the same verified transform Dark Structure Enhance uses), amplifies just the selected band, and rebuilds it exactly. Background Protection holds the effect back over faint sky, where boosting would otherwise amplify noise harder than real structure - measured directly on a real image, which is exactly why it defaults on. Note it boosts STARS as well, since a star is small high-contrast structure by definition (measured: bright star bodies gained +0.105 on a real frame at Strength 1.0) - best run on a starless image, or with Strength kept moderate. Strength starts at 0.0, which is an exact no-op, so nothing happens until you move it.

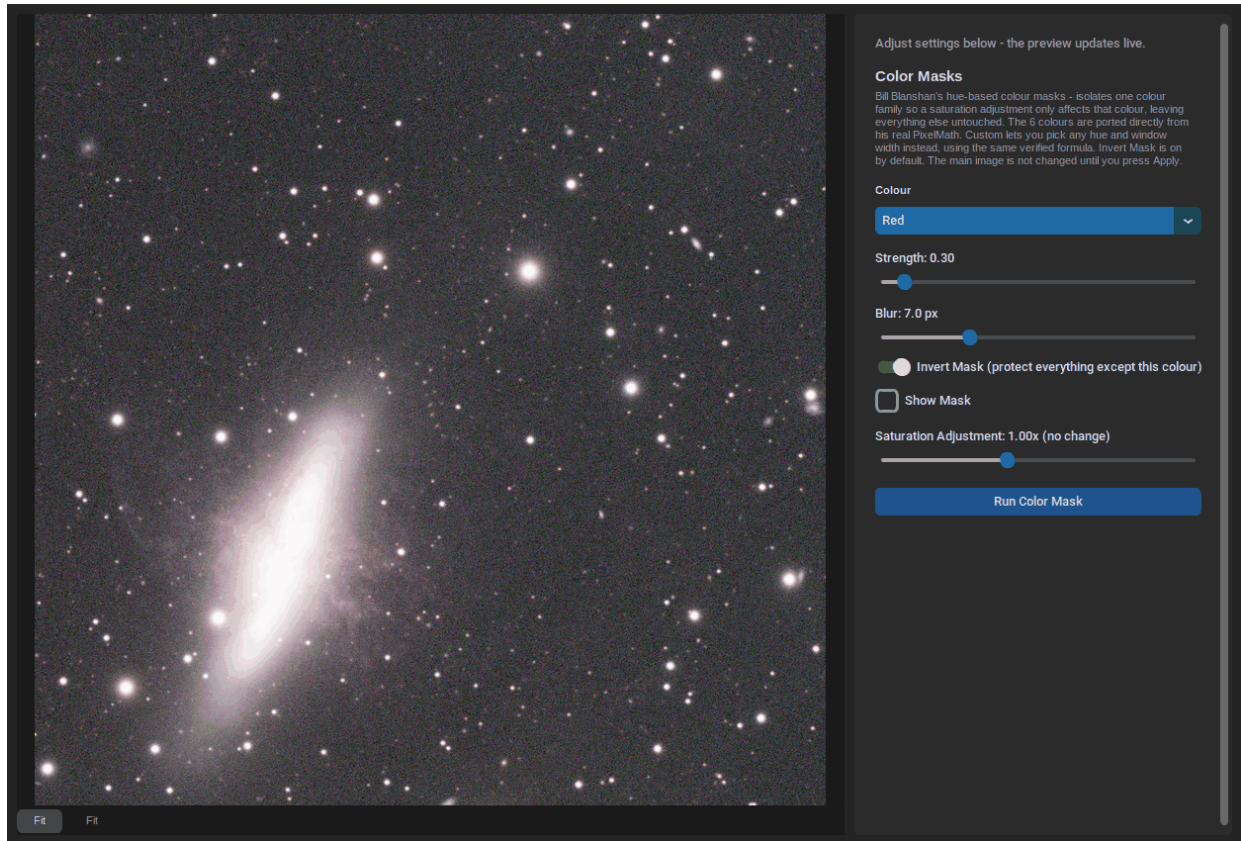
## Colour Tools (5)

## Astro Color Mixer



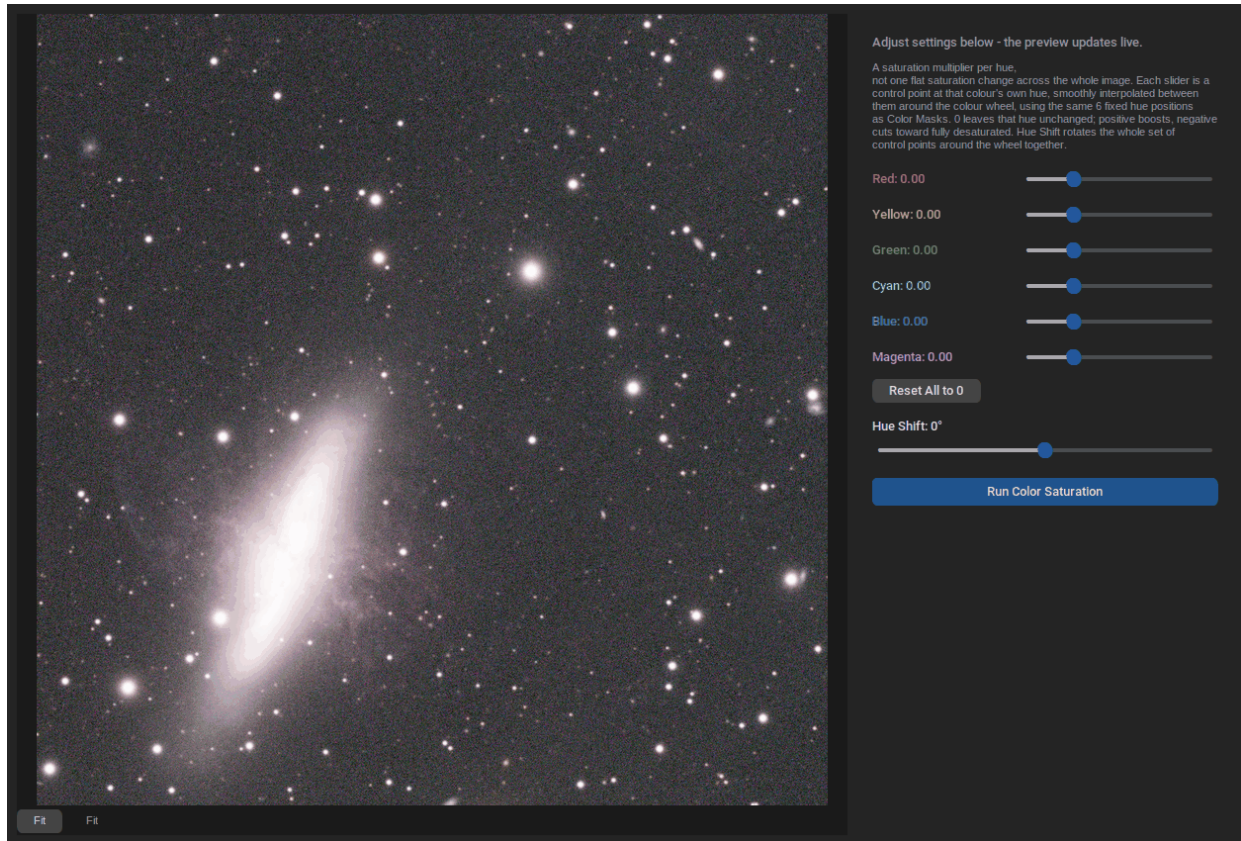
Hue-targeted colour and luminance refinement, adjusting each colour band (Red/Orange/Yellow/Green/Cyan/Blue/Purple/Magenta) independently, with built-in protection for shadows, near-neutral pixels, and highlights/stars. Adapted from Pat Cosgrove's original PixInsight Astro Color Mixer script (Cosgrove's Cosmos), via Yannick Dutertre's (Cuiv the Lazy Geek) Siril port. Only really works on already-stretched (non-linear) RGB data.

## Color Masks



Bill Blanshan's hue-based colour masks - isolates one colour family (Red/Yellow/Green/Cyan/Blue/Magenta) with adjustable strength and blur, so a saturation adjustment only affects that colour, leaving everything else untouched. Custom takes any hue from 0-360 with its own width, and can sample one straight off the image. Show Mask displays what is selected. Invert Mask is on by default.

## Color Saturation



A saturation multiplier per hue (0=unchanged, positive boosts, negative cuts) at each of the 6 fixed hue positions, smoothly interpolated between them, plus a Hue Shift to rotate the whole set together. Uses fixed control points at Color Masks' same 6 hue positions rather than a fully free-form curve.

## SCNR

Adjust settings below - the preview updates live.

Subtractive Chromatic Noise Reduction - corrects a colour cast caused by one channel (usually green) being misaligned with the other two, in a way that protects star colour. Best run on stretched (non-linear) data, after Background Neutralization - running it on an un-neutralized background can destroy the image's colour instead of fixing it.

**Colour to Remove**

Green

**Protection Method**

Average Neutral

Amount: 0.70

☒ Preserve Lightness

Run SCNR

Subtractive Chromatic Noise Reduction - corrects a colour cast from one channel (usually green) being misaligned with the other two, protecting star colour in the process. All 5 protection methods (Average/Maximum/Minimum Neutral, Maximum/Additive Mask), plus Preserve Lightness.

## Correct Magenta Stars

### Correct Magenta Stars

Ready - adjust below, the preview updates live.

Removes the magenta fringing chromatic aberration leaves around stars, usually worst toward the frame corners.

A magenta pixel is one where GREEN sits below both red and blue - the optics focused red and blue slightly differently from green, so the star's edge shows the colour green is missing from. This lifts that deficient green back toward its neighbours - the opposite clamp to SCNR, which pulls an EXCESS channel down.

**Amount: 0.85**

**Threshold: 0.15**

☒ Only correct around stars

☒ Preserve lightness

Reset

Run Correct Magenta Stars

Removes the magenta fringing chromatic aberration leaves around stars - usually worst toward the frame corners, where the optics focus red and blue at slightly different points from green so the star's edge shows the colour green is missing from. It lifts that deficient green back toward its neighbours, which is the OPPOSITE clamp to SCNR: SCNR pulls an excess channel DOWN, this fills a deficient one UP. By default it only touches stars, found with the same detector the palette tools use and grown outward to cover the fringe ring rather than just the core - so magenta nebulosity, which is real signal in narrowband and palette images, is left alone. Turn that off for a broadband image with an overall magenta cast. Luminance-preserving, so removing a fringe does not brighten the star's edge.

## Channels & Palettes (7)

### Split Channels

#### Split Channels

Source: M81\_M82\_RGB.fits

Extracts each channel as its own MONO image, opened in a new Mini Viewer.  
The source is read-only and is never altered.

Useful because each channel can then be processed SEPARATELY by the app's other tools - stretch a weak OIII harder than the Ha, denoise only the noisy one, remove stars from one - and recombined afterwards with Combine LRGB or NB to RGB Stars, which both expect separate mono masters.

For a duo-band one-shot-colour frame: Red is your Ha, Green and Blue carry the OIII.

Mode: Duo-band (Ha + OIII) ▾

Method: Simple (channel split) ▾

OIII from: Green only (better signal-to-noise) ▾

☐ Also extract synthetic Luminance

Luminance from: Equal weight (best signal-to-noise) ▾

Split

Splits the open image into separate MONO images, one per channel, each opening as its own workspace icon - the source is read-only and never altered. Duo-band (Ha + OIII) is the default mode, and Sensor Un-Mix will separate a duo-band OSC frame using its sensor's own filter response rather than just taking the channels as they come. Not needed for palette remapping (PixelMath does any mapping in one expression), but it is the only way to process each channel SEPARATELY: stretch a weak OIII harder than the Ha, denoise only the noisy one, remove stars from one and not the other, then recombine with Combine LRGB or NB to RGB Stars - both of which expect separate mono masters that a one-shot-colour frame could not previously produce inside the app. On a duo-band OSC frame, Red is your Ha and Green/Blue carry the OIII. Optionally also extracts a synthetic Luminance.

## PixelMath

Ready - RGB image. Type an expression to see it applied.

**PixelMath**

Applies an arithmetic expression to every pixel. Use one expression for all three channels, or tick Separate RGB to write one per channel. The preview updates live in the Editor as you type.

Optionally pick extra images below to COMBINE them: refer to them as A, B, C (the channel being evaluated) or A\_R / A\_G / A\_B / A\_L for a specific one. Extra images are read-only, and "Create New Image" opens the result in a new Mini Viewer instead of changing anything you picked.

Image A (not used) ▼

Image B (not used) ▼

Image C (not used) ▼

☐ Separate expression per channel

☐ Rescale result to 0-1

Symbols R G B (open image) L luminance T whole image pi  
 A B C (extra images) A\_R A\_G A\_B A\_L for one channel  
 Maths + - \* / \*\* % ( ) comparisons and or not  
 Functions min max abs sqrt log log10 exp pow mtf iif  
 Stats med mean sdev mad mint maxt - collapse to one number  
 Examples R\*1.2 brighten red  
 (R+G+B)/3 flatten to mono  
 mtf(0.4, R) midtones stretch  
 R - med(R) + 0.5 centre on the median  
 iif(L>0.8, L, 0) keep only the brightest  
 0.8\*A + 0.2\*B blend two open images  
 Separate RGB: B / G / R swaps red and blue  
 Separate RGB: A\_G / 0.8\*A\_R+0.2\*A\_B / A\_B SHO

Apply PixelMath

Create New Image

Expression (all channels) e.g. mtf(0.4, T)

Applies an arithmetic expression to every pixel - one expression for all three channels, or one per channel to mix or swap them. Symbols R, G, B, L (luminance), T (whole image); functions min, max, abs, sqrt, log, exp, pow, mtf, iif; and statistics med, mean, sdev, mad that collapse an image to a single number, so things like 'R - med(R) + 0.5' work as they do in PixInsight. Up to three OTHER images can be loaded into slots A, B and C and referenced as A, B, C (or A\_R, A\_G, A\_B, A\_L for one of its channels), which is how you combine two registered masters in one expression. There is also a Rescale result to 0-1 option for expressions that run out of range.

Live preview in the Editor as you type. Expressions are parsed and whitelisted rather than executed directly, so a typo reports a plain error instead of doing anything unexpected.

## Combine LRGB / Narrowband

### Combine RGB

Select your channel images below.

Combines separate mono master images (e.g. from stacking a mono camera, or extracting individual narrowband filters) into one colour image. An optional Luminance master can be blended in afterwards for extra detail/brightness (LRGB/LSHO etc.), using a separate, often higher-SNR luminance file.

**Mode**

Custom (RGB) ▼

☒ Auto-Align Channels (star-based)

**Red** (no image chosen) ▼

Weight: 1.00

**Green** (no image chosen) ▼

Weight: 1.00

**Blue** (no image chosen) ▼

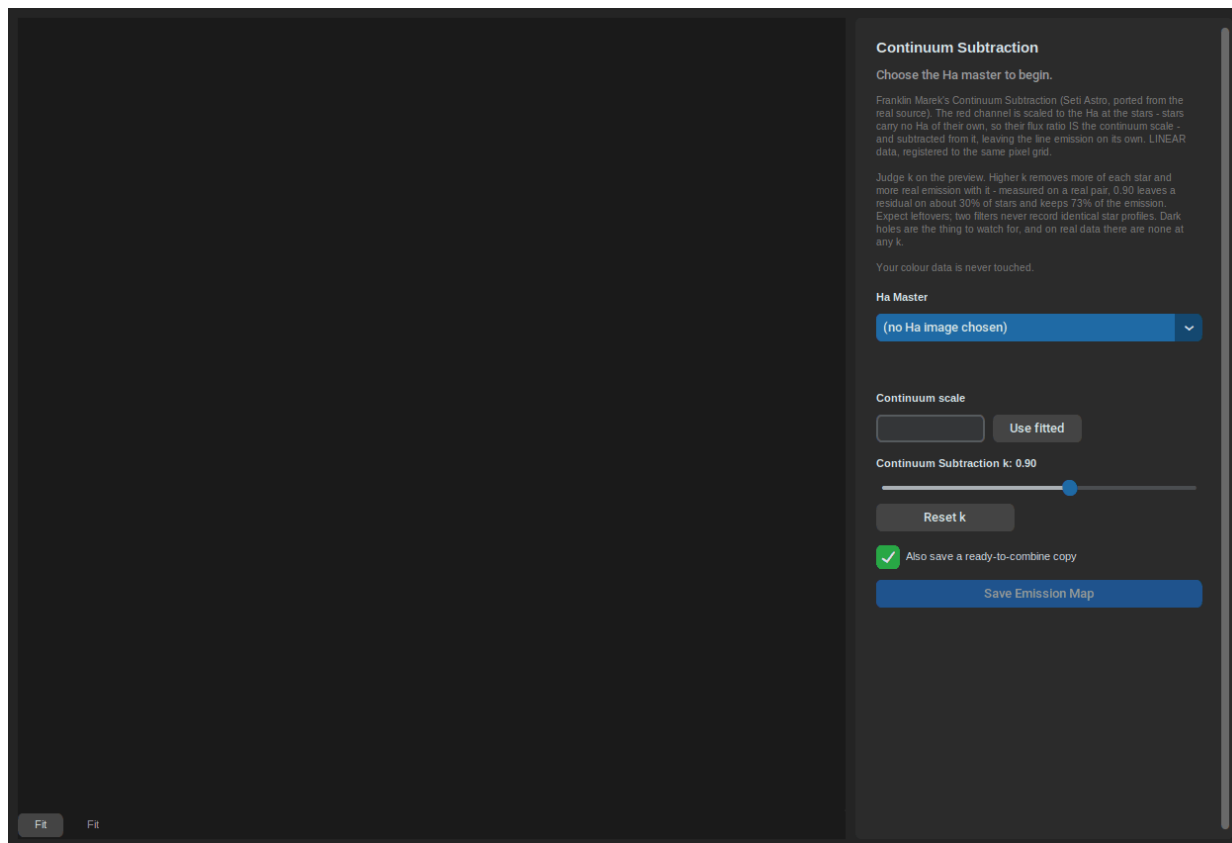
Weight: 1.00

Match Luminance to Colour: 0.00

Run Combine

Combines separate mono master images into one colour image - useful after stacking a mono camera or extracting individual narrowband filters. Custom mode assigns any three images to Red/Green/Blue; HOO/SHO/HSO/HOS presets map your narrowband filters automatically; and Foraxx (adaptive) mixes per pixel from the Ha/OIII ratio rather than mapping channels at all. An optional Luminance master can be blended in for LRGB/LSHO-style results, and Auto-Align Channels registers the masters to each other first if they are not already on the same grid.

## Continuum Subtraction



Strips the starlight out of a registered Ha master and leaves the line emission on its own, as a mono image. Franklin Marek's Continuum Subtraction technique (Seti Astro, ported from the real source): the red channel is scaled to match the Ha at the stars - stars carry no Ha of their own, so their flux ratio IS the continuum scale - and subtracted from it, leaving what the red channel cannot account for.

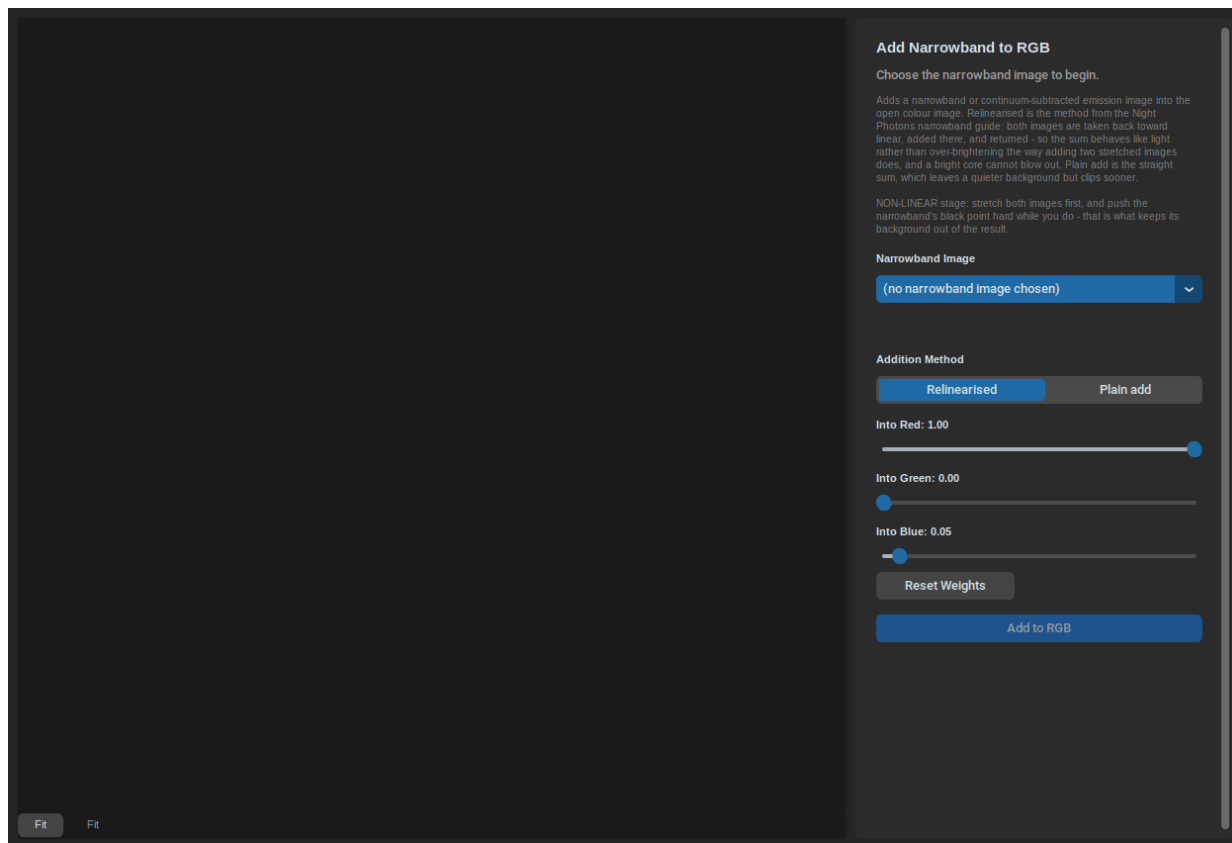
LINEAR-STAGE tool: run it on the linear masters, before stretching, with the Ha registered to the same pixel grid and the colour image open in the Editor.

Judge k on the preview. Raising it removes more of each star and more real emission with it - measured on a real pair, k 0.90 leaves a visible residual on about 30% of stars while keeping 73% of the emission, and k 1.20 leaves 15% of stars for 60% of the emission. Perfect cancellation is not on offer: two filters never record identical star profiles. What k must NOT do is punch dark holes - on real data it does not, at any setting.

The fitted continuum scale is shown and can be typed over - which is what to do on a STARLESS pair, where there are no stars to measure and the fallback estimate is poor: fit on the starry masters, note the number, then enter it here.

Saves TWO images. The linear emission map opens in the Editor - that is the measurement, and its background is meant to look grey. A ready-to-combine copy lands as a workspace icon with its black point already placed from the map's own measured sky and noise; that is the one to pick in Add Narrowband to RGB, and it needs no stretching.

## Add Narrowband to RGB



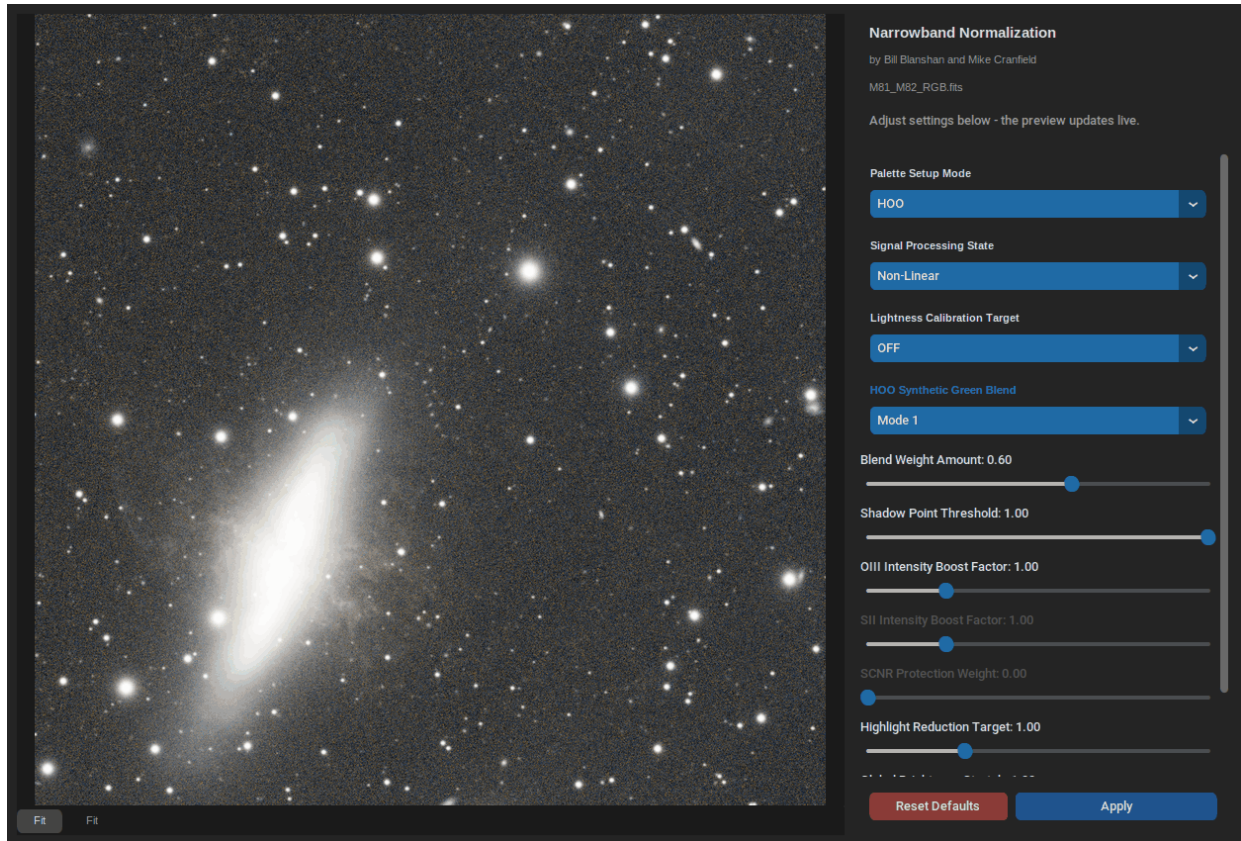
Adds a narrowband or continuum-subtracted emission image into your RGB image, with a weight per channel.

NON-LINEAR stage, and the next step after Continuum Subtraction: the RGB must be stretched, and the narrowband must have its background at zero. Take the ready-to-combine copy Continuum Subtraction saves and that second part is already done - a map whose background is still up is what washes a whole frame red, and no control here can undo it afterwards.

Two methods. Relinearised is the Night Photons guide's: both images are taken back toward linear, added there, and returned, so the sum behaves like light rather than over-brightening, and a bright core cannot blow out. Plain add is the straight sum. With a properly prepared map they measure almost identically at sensible weights; at heavy ones the plain add clips more. Relinearised is the default and the one to reach for. The weights are the same kind of control in both modes, but the same NUMBER does not give the same result - measured on a real pair, 1.00 relinearised took about 0.25 plain to match - so reset and re-judge rather than carrying a value across.

Three weights, defaulting to the guide's own 1.00 red / 0.00 green / 0.05 blue. The small blue is deliberate - it is what makes emission read as pink rather than a flat hard red, and it is the control most worth playing with. All three at zero is an exact no-op.

## Narrowband Normalization



Bill Blanshan and Mike Cranfield's Narrowband Normalization process (HOO/SHO/HSO/HOS) on the currently previewed image. Best run on a starless, background-neutralized image.

## Dual-Band to SHO

Ready - reference OIII/Ha ratio 0.711 (object median 0.842), measured over 34% of the frame.

### Dual-Band to SHO

**SYNTHESIS, not extraction - a duo-band frame contains no sulphur, so this CREATES an SHO-style palette rather than revealing one.**

Turns a duo-band / OSC image into a gold-and-teal SHO-style palette by amplifying how each pixel's OIII/Ha ratio DEVIATES from the field median - the real, measurable variation in the data. Brightness is never invented: every pixel keeps the exact luminance it started with, so this changes colour only. Preview updates live in the Editor; zoom in to judge it properly.

OIII from Average of Green and Blue (recommended) ▾

Colour Separation: 2.60

OIII Reference Point: 25

Gold Limit: 0.34

Teal Limit: 0.42

SII Balance: 0.88

SII Response: 0.85

Green Removal (SCNR): 0.60

Saturation: 1.22

Star Protection: 0.72

Signal Floor: 1.00

Turns a stacked dual-band / OSC colour image into a gold-and-teal SHO-style LOOK. NOT the same job as Dual-Band Extraction on the stacking side, despite the similar name - that one splits your raw CFA subs into genuine Ha and OIII masters before stacking, from real measured photosites. This works on an already-stacked colour image and SYNTHESISES the look: a duo-band frame contains no sulphur at all, so this CREATES a palette rather than revealing one - which is exactly why it is labelled as such. What IS real in the data is the OIII/Ha ratio, and this amplifies how each pixel's ratio DEVIATES from a reference point in the frame's own ratio distribution - the 25th percentile by default, deliberately below the median, since the measured spread is small enough that merely scaling channel levels only tints the frame. Brightness is never invented - every pixel keeps the exact luminance it started with, verified to  $1.8e-07$  - so it changes colour only. Original code, verified on three different real frames. Chains with Narrowband Normalization rather than competing: that tool BALANCES a palette already present, this one creates one.

## CHAPTER 8

# A worked example

---

One dual-band OSC target, from subs to a finished image.

This is one route through the app, not the only one. It is written for a one-shot-colour camera behind a dual-band filter, because that is the case where the most tools come into play.

### 1. Stack it

Load the lights, darks and flats, leave the algorithm on Winsorized Clipping, and stack. What comes out is a linear colour master: almost all of the signal sits close to black, and it will look nearly empty until it is stretched.

### 2. Clean up the background, while it is still linear

Background Extraction or Manual Background Extraction removes the gradient. If the target fills the frame, use the manual one - the automatic tools choose their own sample points, and on a full frame they will happily fit the nebula and then subtract it.

Background Neutralization follows, to take the colour cast out of the sky. A neutral background is a precondition for colour calibration, so this goes before SPCC, not after.

### 3. Calibrate the colour

SPCC plate-solves the image and calibrates against real catalogue photometry. Run it after any cropping - a crop invalidates the plate solution.

### 4. Deal with the stars

If you have StarXTerminator, remove the stars now and keep both halves. Almost everything that follows is better done on a starless image: stretches, structure tools and local contrast all act on stars as well as nebulosity, and stars are the part you least want changed.

### 5. Stretch

Histogram Transformation, GHS or VeraLux, whichever you get on with. Simple Stretch is the one-slider version of the same idea and is enough when you just need the data non-linear so the next tool will work.

### 6. Palette and colour

On a duo-band frame, Dual-Band to SHO synthesises a gold-and-teal look from the OIII/Ha ratio; Narrowband Normalization balances a palette that is already there. They chain rather than compete. Colour Masks, Color Saturation and Astro Color Mixer are the per-hue adjustments, and SCNR takes out a green cast.

### 7. Structure

Multiscale Local Contrast boosts a chosen structure size; Dust Lane Enhancer finds filaments by their shape; Dark Structure Enhance deepens large-scale dark structure. All three are happier on a starless image.

### 8. Put the stars back

Star Stretch the stars-only image, reduce them if they are heavy with Star Reduction, then Screen Stars to recombine. Correct Magenta Stars cleans up chromatic fringing afterwards.

### 9. Finish

Camera RAW Editor for final tone and colour, Crop to taste, then Save As. Keep a 32-bit FITS or XISF of the finished work; export the PNG or JPEG separately, for sharing.

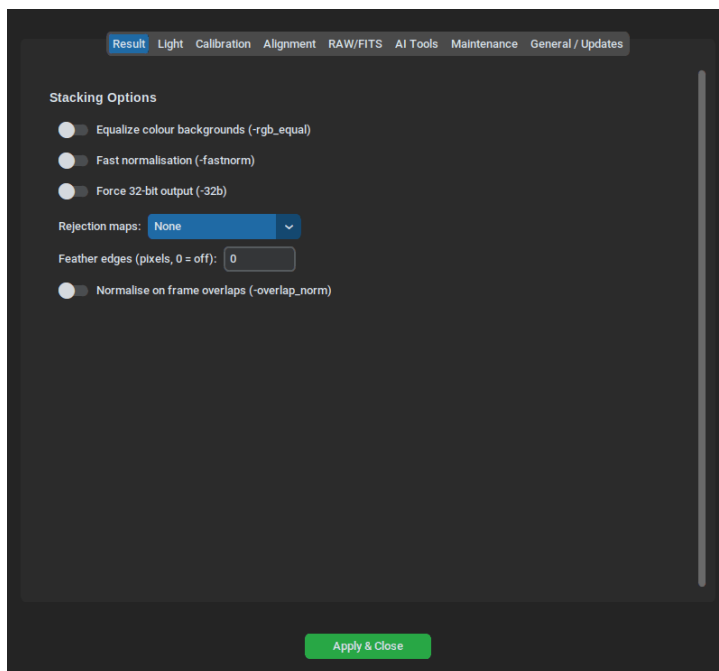
**Note.** Nothing here is compulsory and the order is not fixed. The two orderings that genuinely matter are: background and colour calibration while the data is still linear, and everything else after the stretch.

## CHAPTER 9

# Settings

Advanced Settings, and the housekeeping you occasionally need.

Advanced Settings opens with F4 or from the Settings menu. Each tab scrolls, so the window stays a sensible size however much is on it.



Advanced Settings.

| Tab               | What is on it                                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Result            | What the finished stack is written as, and where.                                                                                         |
| Light             | How light frames are treated through calibration and registration.                                                                        |
| Calibration       | Darks, flats and bias - how they are matched and applied.                                                                                 |
| Alignment         | Registration: how frames are matched to the reference.                                                                                    |
| RAW/FITS          | How raw camera files and FITS are interpreted, including the debayer algorithm for one-shot-colour data.                                  |
| AI Tools          | Where GraXpert and the RC-Astro CLI are installed. Both optional; the tools that need them say so if the path is not set.                 |
| Maintenance       | The scratch folder, the Siril working folder, how many temp sessions to keep, Clear Temp Files, and Reset All Settings.                   |
| General / Updates | App-wide behaviour - remembering the window size and position, warning about unsaved work on exit, interface size - and the update check. |

## Scratch space

Stacking writes a great deal of intermediate data, and so do the tools that produce a second image - split channels, starless results, PixelMath output. It all goes to a scratch folder, and each run gets its own dated session folder

inside it.

Old sessions are cleared at STARTUP, not at shutdown - deliberately, because clearing on exit would destroy unsaved work the moment you closed the app. By default the most recent session is kept and everything older is removed; Maintenance sets how many to keep. Clear Temp Files empties the lot, now, as the manual escape hatch.

**Note.** Both the scratch folder and the Siril working folder can be moved to another drive in Maintenance, which is worth doing if your system drive is small. The app write-tests the new location before accepting it.

## Interface size

Adjust interface size, on General / Updates, scales the whole interface RELATIVE to Windows' own display scaling - so on a 4K screen already running at 125%, setting 80% here lands you at true 1:1. It needs a restart to take effect.

## Updates

Check for Updates compares the newest release on GitHub against the version you are running. If there is one, a green notice appears under the About / Credits button in the sidebar and stays there until you act on it. The check on startup is off by default.

An update downloads the new release, waits for the app to close, swaps the program folder and relaunches - and puts the old folder back if anything goes wrong.

## CHAPTER 10

# If something goes wrong

The console first, then this list.

The console at the bottom of the window records everything the app and the engine do. Clear Log before running the thing that is misbehaving, then Copy Log afterwards, and you have exactly that run and nothing else.

| Symptom                                                       | Most likely cause                                                                                                               |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| A tool appears to do nothing                                  | A mask is active and the area you are looking at is black in it. Check the mask dropdown on the editor toolbar.                 |
| A tool refuses to open and the console says a path is not set | GraXpert or the RC-Astro CLI is not installed, or its path is not filled in on Advanced Settings > AI Tools.                    |
| Stacking stops part-way through                               | Siril aborts a script at its FIRST error, so the cause is in the console lines just before it stopped, not at the end.          |
| Registration finds too few stars                              | Lower the Star Detection Threshold. On a sparse field it can be the difference between registering and not.                     |
| A stretch tool behaves oddly, or barely at all                | The data is still linear. Tools marked as working on non-linear data need a stretch first - Simple Stretch is enough.           |
| Colour calibration or plate solving fails                     | No internet connection, or the image has been cropped since it was solved. Crop first, then solve.                              |
| The disk fills up                                             | Old scratch sessions. Clear Temp Files in Maintenance, and consider moving the scratch folder to a larger drive.                |
| Antivirus flags the .exe                                      | PyInstaller's bootloader is a known false-positive pattern and the binary is unsigned. Add an exclusion for the program folder. |
| The window opens off-screen                                   | It does not - a saved position on a monitor that is no longer connected is detected and the window falls back to maximised.     |

## CHAPTER 11

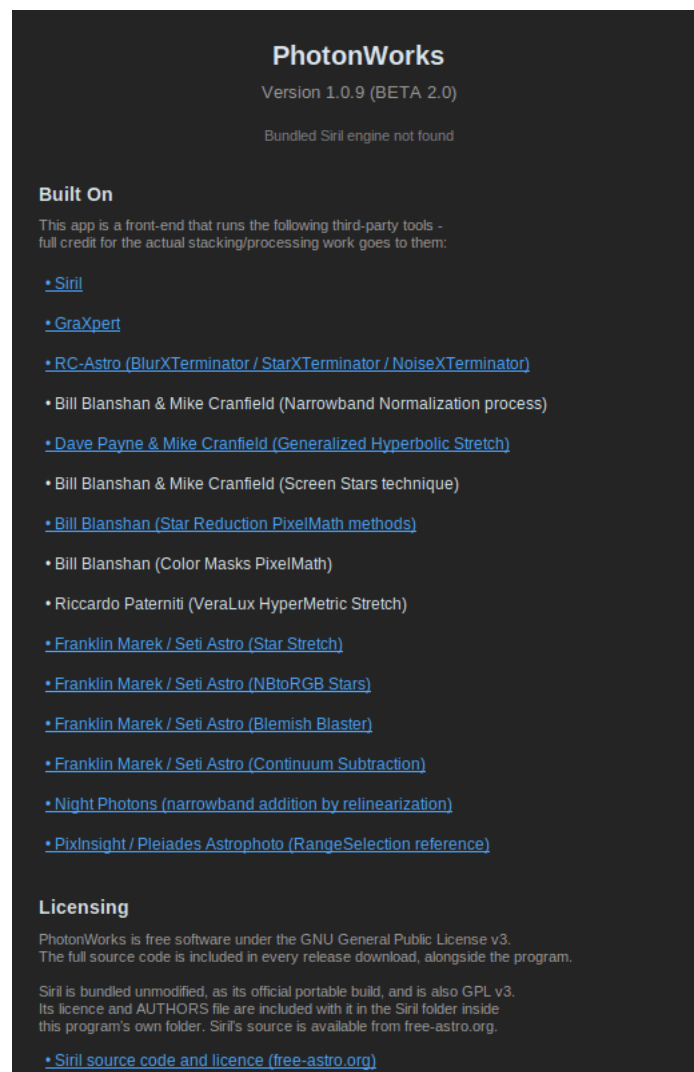
# Credits and licensing

---

Who did the work the app is built on.

PhotonWorks is a front-end. The stacking engine, several of the processing techniques, and two of the optional tools are other people's work, used with credit. The app's own About dialog carries this same list, with links.

|                                                                                      |
|--------------------------------------------------------------------------------------|
| Siril                                                                                |
| Siril Gaia DR3 Astrometry Catalogue - Richard, Melis, Knagg-Baugh & Cass (CC BY 4.0) |
| GraXpert                                                                             |
| RC-Astro (BlurXTerminator / StarXTerminator / NoiseXTerminator)                      |
| Bill Blanshan & Mike Cranfield (Narrowband Normalization process)                    |
| Dave Payne & Mike Cranfield (Generalized Hyperbolic Stretch)                         |
| Bill Blanshan & Mike Cranfield (Screen Stars technique)                              |
| Bill Blanshan (Star Reduction PixelMath methods)                                     |
| Bill Blanshan (Color Masks PixelMath)                                                |
| Riccardo Paterniti (VeraLux HyperMetric Stretch)                                     |
| Franklin Marek / Seti Astro (Star Stretch)                                           |
| Franklin Marek / Seti Astro (NBtoRGB Stars)                                          |
| Franklin Marek / Seti Astro (Blemish Blaster)                                        |
| Franklin Marek / Seti Astro (Continuum Subtraction)                                  |
| Night Photons (narrowband addition by relinearization)                               |
| PixInsight / Pleiades Astrophoto (RangeSelection reference)                          |



About / Credits, inside the app.

## Licence

PhotonWorks is released under the GNU General Public License version 3. That is required rather than chosen: the app contains faithful ports of GPL-licensed originals, so copyleft applies to the combined work.

In practice that means the source is available alongside every release, you may study and modify it, and anything you distribute that is built from it carries the same licence and the same source obligation. It also means the program comes with no warranty of any kind.

The bundled Siril carries its own GPL v3 licence text and its AUTHORS file inside the program folder. Neither should be removed.

**APPENDIX A**

# Menus and keyboard shortcuts

---

Taken from the app's own menu definitions.

**File**

|                        |              |
|------------------------|--------------|
| Open Image...          | Ctrl+O       |
| Smart Import Files...  |              |
| Smart Import Folder... |              |
| Save Image             | Ctrl+S       |
| Save Image As...       | Ctrl+Shift+S |
| Open Output Folder     |              |
| Clear Frame List       |              |
| Exit                   | Alt+F4       |

**Edit**

|                            |        |
|----------------------------|--------|
| Undo                       | Ctrl+Z |
| Redo                       | Ctrl+Y |
| Refresh Metrics            | F5     |
| Hide / Show Left Panel     | F9     |
| Hide / Show Info + Console | F8     |
| Clear Console              |        |

**Settings**

|                        |           |
|------------------------|-----------|
| Advanced Settings      | F4 \u25b8 |
| Clear Temp Files       |           |
| Clear Siril Work Files |           |

**Window**

|                 |
|-----------------|
| Tile Windows    |
| Cascade Windows |
| No images open  |

**Help**

|                       |
|-----------------------|
| How Editing Works     |
| What's New            |
| Check Engine Versions |

About

|                   |    |
|-------------------|----|
| About PhotonWorks | F1 |
|-------------------|----|

The Tools menu is generated from the same list as the sidebar, so it always holds exactly the tools in Chapter 7, in the same groups.

## APPENDIX B

# Version history

The three most recent releases. The app's own What's New, under the Help menu, carries the whole history, and so does CHANGELOG.md in the release.

## v1.0.9

FIXED: the startup splash jumped up and to the left and shrank a moment before the main window appeared, on any display not set to 100%

## v1.0.8

SEVERAL IMAGES OPEN AT ONCE, each in its own window with a title bar you can drag and a corner you can resize

Added TILE and CASCADE to the Editor toolbar, re-fitting each image to its new window

Added a WINDOW MENU between Settings and Help, listing every open image - minimised ones included - with the active one ticked

Added KEYBOARD WINDOW CYCLING: Ctrl+Tab and Ctrl+Shift+Tab, with Ctrl+F6 and Ctrl+Shift+F6 doing the same

Added a 1:1 button to each image window; the toolbar's 1:1 and Fit are now global to match

Clicking an image window makes it the one the toolbar, tools, Undo, Save and readout all point at

Picking several files in Open... now shows them all, tiled, instead of one in the Editor and the rest as icons

Double-clicking a workspace icon now joins the images already on screen instead of replacing them

Added a SPLASH SCREEN to the built .exe, showing the logo as soon as the icon is double-clicked

Renamed Clear Editor to CLEAR WORKSPACE - it closes every open image, after a warning listing what would be lost

The empty Image Editor now offers drag & drop like the Files Viewer, and only when drag & drop is actually working

Dropping a workspace icon onto a Mini Viewer now MOVES the image rather than copying it

Opening an icon whose image is sitting in a Mini Viewer now clears that Mini Viewer, so the image is in one place

Open, Save, Output Folder, Auto-open Result, Tile and Cascade are white on the top toolbar again rather than orange

Image window title-bar buttons are bigger and held clear of the resize corner

Drag & drop now reports in the console whether it is working, instead of failing silently

ONE PICTURE PER IMAGE: every window draws the same full-resolution picture, so switching windows is a redraw rather than a two-second rebuild

A mono image is stretched once rather than three times - 2.77s against 5.52s on a 24 megapixel frame

Image pictures are built one at a time, in order, and the image you click goes to the front of the queue

The rendered-image cache is sized to the workspace instead of being fixed at four

The History panel is no longer rebuilt while it is collapsed

Selecting an image no longer forces three full repaints of the application

Minimising the image you are editing hands over in the same turn of the event loop

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
| A parked image's icon is made from that image's own picture rather than read from the file again                             |
| The screen stretch belongs to the IMAGE: turning STF off, or switching Linked/Unlinked, no longer disturbs the other windows |
| Image windows take the shape of the image rather than of the space they are given                                            |
| Every open image is read by the same loader the editor uses, so float data is normalised the same way                        |
| A window you are not editing is drawn with the same stretch curve as one you are                                             |
| FIXED: dragging an image window was slow, left a trail of half-drawn edges, and smeared blue streaks behind it               |
| FIXED: clicking between image windows blanked the one you had just left, and flickered                                       |
| FIXED: minimising one window disturbed every other window, or blanked the Editor                                             |
| FIXED: closing or minimising the last image left an empty window standing where it had been                                  |
| FIXED: SWAP left the window still describing the image that had just left it                                                 |
| FIXED: the wheel zoomed the wrong image, and hovering a window reported another image's pixels                               |
| FIXED: a window could open larger than the Editor, or slide into its own corner after a Tile or Cascade                      |
| FIXED: opening an image while others were in windows often showed its name but no picture                                    |
| FIXED: opening several images at once painted several arrangements before settling on the right one                          |
| FIXED: the empty Image Editor came up completely blank on a fresh start                                                      |
| FIXED: the empty-workspace logo appeared in the top-left corner for a moment before centring                                 |
| FIXED: dropping an icon where two Mini Viewers overlap loaded it into the one underneath                                     |
| FIXED: the Camera RAW Editor's live preview failed on every mono image                                                       |
| FIXED: an XISF window did not take its image's shape, leaving black bands beside the picture                                 |
| FIXED: an image window shaped itself from a downscaled copy, and a saved file left the window showing the old version        |
| FIXED: the picture in every window jumped three pixels for a single frame on each switch                                     |
| FIXED: dragging a file onto the editor stopped working over the workspace icons                                              |
| GROUNDWORK for the multi-window workspace: each image has its own canvas and its own frame                                   |

## v1.0.7

|                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|
| The crop box TILTS - drag the grip outside its top-right corner and Apply Crop hands the tilted rectangle back upright |
| A tilted crop keeps a plate solution pointing at the same sky, verified against astropy                                |
| A tilted crop is refused on a still-Bayer file, where it would blend the colour filter pixels together                 |
| The crop box has eight handles, four corners and four sides, and the opposite one stays put while you drag             |
| REMOVED: the green stippled wash inside the crop box - it covered the detail you are framing                           |
| Set Preview Region has the same eight handles, and its overlay no longer flickers while dragging                       |
| Added MINIMISE, FIT and CLOSE in the top-right corner of the Editor; Minimise parks the image as a workspace icon      |
| Every tool has its own icon, in the sidebar, in the menu and on its own window's title bar and taskbar button          |

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| Blemish Blaster spots stay editable - click to select, drag to move or resize, Delete removes just that one        |
| Manual Background Extraction's exclusion areas are editable, with Backspace to take back the last corner           |
| PICK THE MASK LIMITS OFF THE IMAGE: RangeSelection's Lower and Upper can be set by clicking your own sky or nebula |
| Background Neutralization has Pick Background - one click on your sky brackets it                                  |
| LinearFit has Pick Reject Low and Pick Reject High, set by clicking what you want kept                             |
| Curves can pick from the image - a click marks the value under the cursor, a double-click anchors the curve there  |
| Renamed Narrowband Addition to ADD NARROWBAND TO RGB, with a Plain add mode alongside the relinearised one         |
| Add Narrowband to RGB warns when the two images would lift the background rather than add emission                 |
| Continuum Subtraction saves a second, ready-to-combine copy with its black point already placed                    |
| Continuum Subtraction's continuum scale can be typed over the fitted one, with Use fitted to put it back           |
| LinearFit's reference is chosen with one dropdown - workspace images, then Mini Viewers, then Browse               |
| The hover readout shows K for mono images, and Preview Region gives its median and mean                            |
| The magnifier has a crosshair and a sample box, showing the single pixel and the box median together               |
| RangeSelection's Smoothness is now called BLUR, which is what it is, and reaches 80 instead of 10                  |
| Adjusting a RangeSelection mask while it is Active is no longer slow - 2.7 seconds a slider move, gone             |
| Large mask blurs are built at reduced resolution, which the result cannot tell apart                               |
| FIXED: Undo after LinearFit showed a blown-white image until the stretch was toggled                               |
| FIXED: Ctrl+Z while typing in a box undid the image, and reached past a tool window to the image behind            |
| FIXED: closing the last workspace icon left the canvas still telling you to double-click one                       |
| FIXED: the View FITS Header window kept the plain application icon and never got the dark title bar                |
| FIXED: a crash when closing a tool dialog - the viewer was asked to re-render with no file given                   |
| FIXED: cancelling Save As New discarded the result instead of leaving it in the editor                             |
| FIXED: renaming an image did not follow it onto a workspace icon, which came back with a generic name              |